NameRank: Measuring LLM-Mediated Recognition as the Post-Bibliometric Impact Channel

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Abstract

Li [2026] found that bibliometrics explain only ∼35% of whether a frontier model recognizes a researcher, the residual driven by name uniqueness, named-artifact attachment, and subfield density. We treat that residual as the measurand. **NameRank** is a continuous $[0, 1]$ recognition score: we probe a panel of $M = 37$ frontier models with one open-ended question per entity ("tell me what you know about [name], who [context]") and score each response on a multiplicative coverage×accuracy axis with an independent LLM judge against a curated gold answer. Across 5,719 entities in 54 cohorts (211,603 records, plus an 8,880-record Chinese sub-run), four findings stand out. **(i) The credential treadmill**: 7 of 9 Olympic-style credentials (IMO, IOI, CMO, Rhodes, MSRA) score *at or below* a long-tail working-researcher baseline—the credential no longer propagates a name, named post-credential production does. **(ii) Artifact>creator inversion**: in 8 of 11 verified (creator, artifact) pairs the tool out-ranks its maker, and a context-injection experiment corroborates the underlying attribution mechanism. **(iii)** ***h*-index dominates raw citations** (within-cohort R^2=0.40 vs. 0.14, 10-fold CV R^2=0.38; citations add zero marginal value), because NameRank rewards name-recurrence across distinct works. **(iv) A corpus-density gradient**: Stanford-affiliated CS faculty score ∼2× Tsinghua-affiliated (0.54 vs. 0.26), part of a country-level English-text ladder (institution explains ∼16% of within-cohort variance by adjusted R^2/ICC; the naive 54% is inflated by 320 near-singleton institution categories). The metric is largely *silent rather than misleading* below threshold—a contemporaneously-published author cohort scores 0.042, read against a calibrated synthetic-null floor—and a battery of robustness checks (paraphrase, training-cutoff, three-judge, synthetic-null, confound controls) confirms the findings are not artifacts of wording, model vintage, judge family, or gold-answer length. We release all probes, gold answers, responses, and code: `https://github.com/19PINE-AI/namerank`.

1 Introduction

Li [2026] reports an empirical finding inside an instrument built for a different purpose. The Incompressible Knowledge Probes (IKP) project measures effective knowledge capacity of frontier language models by counting which rare facts they have absorbed; on the way to that calibration, the 345-researcher subprobe was found to correlate only modestly with citation count (Pearson $r = 0.59$, $R^2 \approx 0.35$). The other 65% of recognition variance, the paper showed, decomposed in a controlled 2×2×2 audit into *name uniqueness*, *named-artifact amplification*, and *subfield-ecosystem density*. The framing was descriptive: recognition appeared as a side-product of the calibration ladder, characterized but not measured as a primary variable.

We argue here that the 65% residual is the measurand, and we build a continuous instrument for it.

The substantive motivation is that the channel through which human curiosity is mediated is changing. A practitioner deciding whether to read a paper, hire a candidate, attend a talk, or invest in a startup increasingly arrives at the question with whatever a frontier language model says about the entity, often before any human source intervenes. We do not measure the prevalence of this LLM-first lookup behavior—it is a usage trend we take as a premise, not a result—but conditional on it mattering, what the model says is determined by whether the entity's name has propagated into the training corpus. If the corpus has absorbed the name with a clean attribution chain, the practitioner is presented with a coherent summary; if it has not, the model says "unknown" or, worse, fabricates a plausible-sounding bio. The information surface that mattered ten years ago—Google rankings, conference plenaries, paper citation counts—has been re-mediated by a small number of frontier models, and the question of whether an entity's name has propagated into those models' weights is now itself a substantive measurement problem.

A normative grounding makes the operationalization sharper. The IKP author's competition-math cohort circulated, for years, the following definition due to Jiayi Weng (an OpenAI infrastructure engineer and author of the Tianshou reinforcement-learning library):

> *rúguǒ rénshēng shì yóuxì, fēnshù jiùshì jìzhù nǐ míngzi de rénshù*
> *—If life is a game, the score is the number of people who remember your name.*

In the LLM era this suggests a heuristic factorization (not an identity we estimate—the second factor is left unmeasured):

$$\text{Recognition impact} \approx \underbrace{\text{NameRank}}_{\text{per-LLM presence}} \times \underbrace{\text{Deployment surface}}_{\text{humans querying that LLM}}$$

To the extent the deployment surface is approximately constant across well-known entities at the relevant time horizon, the first factor carries the cross-entity variation. That first factor is what we measure. We do not claim NameRank measures *quality* or *merit*; we measure whether the corpus that mediates human curiosity at scale has internalized the entity's name. That is what "remember" means in the LLM era.

Why a new metric rather than reading IKP Sec. 5.7 off the shelf. IKP's recognition section was constructed as a side-product of a 6-landmark tier ladder, with a binary CORRECT/WRONG verdict per probe and a 4-way evidence-aware judge for researcher probes specifically. The resulting recognition scores live on a tier scale (S/A/B/C/D/F) at population level and a $[0,1]$ per-probe rate at individual level; they are calibrated for parameter estimation, not for cross-entity comparison. Three properties needed for an impact metric are absent: (i) a continuous score that places industry founders, OSS maintainers, podcast hosts, mid-career CS faculty, and Walmart CTOs on the same axis; (ii) a hallucination-resistant scoring rule that distinguishes "confident bio of the right person" from "confident bio about the wrong person on the same topic"; (iii) explicit cross-model variance as a separate, retained signal rather than absorbed into a tier label.

NameRank provides these. The probe is open-ended (not closed-factoid), the score is multiplicative coverage$\times$accuracy by an LLM judge against a 100–200 word gold answer, and we retain per-model scores so within-entity disagreement is itself a measurand. We evaluate 5,719 entities across 54 cohorts—credentialed individuals, working academics, founders, named artifacts, mid-tier cross-profession figures, and diagnostic cases—against a panel of $M = 37$ frontier models, yielding 211,603 probe records.

Contributions. (Numbers are deferred to the cited results sections.)

1. A **continuous cross-domain recognition instrument** that places researchers, founders, OSS authors, public websites, and named artifacts on one $[0,1]$ axis—no prior instrument does (Section 6.1).
2. The **credential-treadmill thesis**: most Olympic-style intellectual credentials (IMO/IOI/CMO/Rhodes/MSRA) score at or below the working-researcher baseline; the credential no longer propagates a name, named post-credential production does (Section 6.2).
3. The **artifact$>$creator inversion**, measured observationally and corroborated by a context-injection experiment, traced to attribution-direction asymmetry (Section 6.3).
4. Refined external validity—***h*-index dominates raw citations** because NameRank rewards name-recurrence across distinct works, not attribution-per-paper—and a **corpus-density and cross-language gradient** driven by English-text volume per affiliation (Sections 6.4–6.5).
5. A full **methodological validation**—multiplicative LLM-judge scoring, variance decomposition, and paraphrase, training-cutoff, three-judge, synthetic-null, and confound controls (Section 6.7)—with an **open release** of all probes, gold answers, 211,603 responses, and code: `https://github.com/19PINE-AI/namerank`.

Figure 1 previews the central result—the continuous cross-domain placement that no prior instrument supports.

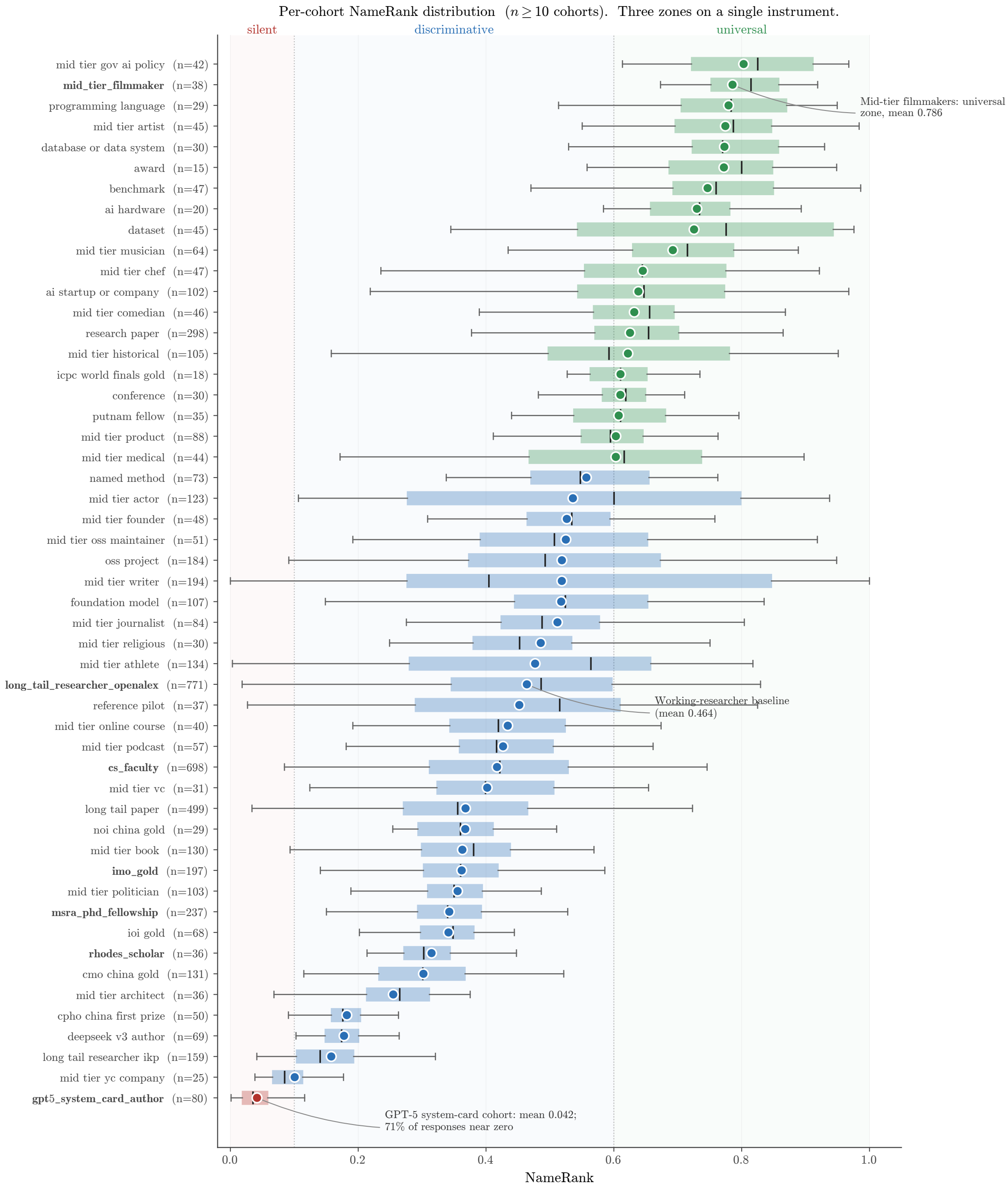

Figure 1. Per-cohort NameRank distribution. Each box is a cohort ($n \geq 10$); the dot is the cohort mean, the box covers the IQR, and the inner line is the median. Three zones are visible on the same axis: *silent* (mean ≤ 0.1, dominated by the GPT-5 system-card author cohort at 0.042 with 71% of responses scoring near zero, in red); *discriminative* (0.3–0.6, CS faculty 0.417 and OpenAlex long-tail 0.464, in blue); *universal* (> 0.7, named artifacts and high-profile mid-tier figures, in green). The credential cohorts (IMO/IOI/CMO/Rhodes/MSRA, bold-faced labels) cluster in the lower half of the discriminative zone, below the OpenAlex working-researcher baseline (Section 6.2).

2 Background and Related Work

2.1 Recognition as a Side-Product of IKP Sec. 5.7

The direct predecessor of this work is Li [2026, Sec. 5.7]. That paper's recognition analysis showed three things relevant here. First, on 345 CS-researcher probes scored by 140 models, log(citations) explains only $\sim$35% of recognition variance, with the remainder attributed in a controlled $2\times2\times2$ audit to *named-artifact amplification* (e.g., FlashAttention/Tri Dao, IPFS/Psaras), *name uniqueness* (common East Asian surnames attenuate recognition by $\sim2\times$), and *subfield-ecosystem density* (ML researchers floor at 43% recognition; Systems researchers can fall arbitrarily low). Second, on 557 Wikidata-grounded entity probes, pageviews dominate sitelinks in joint regression, and domain-specific multipliers (journal founding years $+0.20$, place founding years -0.31) trace to the *structure of web discourse around each fact type* rather than entity prominence. Third, recognition was framed as a benchmark byproduct correlated with bibliometrics rather than a measurand in its own right.

We treat the third point as a methodological choice rather than a settled question, and the first two as foundational mechanisms whose operationalization the present paper completes. The substantive differences from IKP Sec. 5.7: a continuous $[0, 1]$ score over a 37-model panel (vs. a 6-landmark S/A/B/C/D/F tier ladder); 5,719 entities across 54 cohorts spanning industry, OSS, and mid-tier figures (vs. 345 CS-researcher and 557 Wikidata probes); open-ended probes scored on multiplicative coverage$\times$accuracy (vs. closed-factoid binary verdicts); h-index rather than raw citations as the bibliometric correlate; and cross-model variance retained as signal rather than absorbed into a tier label.

2.2 Scientometric Measures of Researcher Impact

The classical instruments for researcher impact—citation count, h-index [Hirsch, 2005], journal impact factor, g-index, i10-index—measure intellectual reach within the academic publication system. They are well-defined, well-instrumented, and increasingly understood as failing under two pressures relevant here. Larivière et al. [2019] document inflation of authorship lists and uneven attribution. The *hyperauthorship* problem [Cronin, 2001] captures the difficulty of pricing individual contributions to thousand-author papers (frontier-LLM technical reports being the canonical example): individual citation count for a co-author of GPT-4 is mechanically large but tells us almost nothing about that author's name-recognition. Wapman et al. [2022] document strong geographic and prestige-hierarchy effects that bias citation-based metrics by an order of magnitude across institutional tiers. Sinatra et al. [2016] introduce the Q parameter to separate productivity from typical-paper-impact, partially addressing within-researcher variance.

None of these instruments cross the academic boundary cleanly. Citation count does not measure Sam Altman, Aravind Srinivas, or a Walmart CTO; h-index does not measure the creator of nanoGPT. Bespoke domain metrics (Google PageRank for websites, GitHub stars for OSS, Spotify monthly listeners for musicians, IMDb scores for film) do not interconvert. The cross-domain measurement of eminence has a long prior tradition in *historiometry* [Simonton, 1999], which quantifies the reputational standing of eminent individuals from archival traces; NameRank can be read as a historiometric instrument whose archive is the frontier-model training corpus rather than biographical dictionaries and citation indices. For impact assessment that spans academia, industry, OSS, and the long tail of solo operators, a cross-domain instrument is needed; NameRank is the first we are aware of that reads this archive directly.

2.3 LLM-as-Judge for Quality and Impact

Thelwall [2024, 2025b,a] use ChatGPT to score paper *quality* against the UK REF rubric, finding moderate alignment with human reviewers after multi-iteration averaging within a single model. Kousha and Thelwall [2025] extend to societal-influence assessment. Liu et al. [2026] forecast citation counts via LLM, with citations as the target variable. These differ from NameRank in two respects. (i) They ask the LLM to *judge* quality; we ask what the model has *absorbed* about the entity. (ii) They average within a single model across iterations; we average across 37 models in a single pass, treating cross-model variance as informative rather than as noise to be averaged out.

The known LLM bias most relevant to our setting is the *Matthew effect* [Merton, 1968] as it propagates through LLM training corpora: frontier models recall already-highly-cited work disproportionately, inflating the rich-get-richer dynamic familiar from the bibliometrics literature. We inherit this bias and treat it as a known property of the measurand—NameRank measures *what frontier models have absorbed*, which is itself shaped by the Matthew dynamic. The bias is part of what we measure, not a flaw to be corrected.

2.4 Knowledge Storage and Retrieval Mechanisms

The mechanism by which a model's training corpus shapes its open-ended recall is well-studied. Transformer feed-forward layers function as key-value memories for factual associations [Geva et al., 2021], with specific "knowledge neurons" responsible for individual facts [Dai et al., 2022, Meng et al., 2022]. Petroni et al. [2019] establish that pretrained models serve as implicit knowledge bases; Roberts et al. [2020] that model size directly determines factual capacity. Kandpal et al. [2023] show that accuracy on rare facts scales log-linearly with model size, and Mallen et al. [2023] that LLMs are unreliable on less popular facts. The *Law of Knowledge Overshadowing* [Zhang et al., 2025] shows that popular knowledge suppresses less popular knowledge, with hallucination rate increasing log-linearly with knowledge popularity. All these underlie our threshold-cleanliness property: NameRank reports near-zero for entities below the corpus-presence threshold rather than producing noisy rankings, because the underlying retrieval mechanism is itself thresholded.

2.5 Sociology of Recognition and Career Outcomes

The credential-decay finding (Section 6.2) connects to a body of work on long-run educational and prize outcomes. Lubinski et al. [2020] (the Study of Mathematically Precocious Youth, 50-year longitudinal) find that early-life mathematical talent strongly predicts later professional and creative outcomes; Agarwal and Gaule [2020] (a Bell Labs-style natural experiment via Math Olympiad medal cutoffs) find that IMO medals causally increase subsequent academic productivity. Both use academic-publication outcomes as the dependent variable. Güllich et al. [2025] (extending to Olympic athletes and Nobel laureates) document substantial within-elite heterogeneity. Our finding is orthogonal: when the dependent variable is LLM-mediated name recognition rather than academic publication, the once-prestigious credentials we measure (IMO, IOI, CMO, Rhodes, MSRA Fellowship) explain less of the outcome than h-index. This does not refute the academic-outcome literature; it identifies a different post-credential propagation channel. In the language of signaling theory [Spence, 1973], a credential is a costly signal whose value depends on the channel that carries it to receivers; the credential treadmill (Section 6.2) is the observation that the LLM-corpus channel does not transmit the credential itself, only the named, indexable production that may follow it.

3 The NameRank Construct

3.1 Definition

For an entity e with a disambiguating context $c(e)$ and a curated gold answer $g(e)$, and a model panel $\mathscr{M}$ of size $M = 37$, NameRank is

$$\text{NameRank}(e) \;=\; \frac{1}{M}\sum_{m\in\mathscr{M}} \text{cov}(r_{e,m}, g(e)) \cdot \text{acc}(r_{e,m}, g(e)), \tag{1}$$

where $r_{e,m}$ is the response of model m to the open-ended probe

> "*Tell me what you know about [name], who [context]. If you do not recognize this entity, respond with 'unknown'. Limit your response to about 150 words.*"

and $\text{cov}, \text{acc} \in [0,1]$ are independent coverage and accuracy scores produced by an LLM judge (Section 4.3). The multiplication is critical: a fluent-hallucination response with high topical coverage but factually wrong specifics has $\text{cov} \to 1$ but $\text{acc} \to 0$, yielding NameRank contribution $\to 0$ for that (entity, model) pair. Appendix H gives three worked (entity, model, response) triples spanning the score range to make the multiplicative rule concrete.

3.2 Three Properties

1. Cross-domain unity. Researchers, founders, OSS authors, podcast hosts, comedians, and CTOs map onto the same $[0,1]$ scale. We demonstrate this empirically (Figure 1).

2. Threshold cleanliness. Above an elite-recognition threshold (roughly: corpus-presence density crossing the model retrieval threshold), NameRank is informative across domains. Below the threshold it is *largely silent rather than misleading*—for cohorts with dense, specific gold answers it does not produce spurious rankings of unrecognized entities. This cleanliness is gold-dependent, not absolute: cohorts whose gold answer is short boilerplate (e.g. the olympiad credentials, whose gold is year + country + medal) carry an elevated floor near ~ 0.20 that a model can partly satisfy from the disambiguating context alone, so for those cohorts the metric is silent only above that floor (Section 6.7.7). The GPT-5 system-card author cohort ($n = 80$, paper published days before training cutoff for most models)—a dense-gold cohort—instantiates the clean case: mean NameRank 0.042, with 58% of responses outright refusing ("unknown") and an additional 13% scoring near zero on the multiplicative axis (71% silent overall). The metric correctly reports "I have no information about this entity" rather than producing noisy fractional scores. We show in Section 6.7.5 that this silence is *intrinsic* rather than a corpus-timing artifact: it persists even in models whose training cutoff postdates the cohort's emergence and that therefore demonstrably ingested the source document—presence in the training corpus is necessary but not sufficient for recognition. A synthetic-null calibration (30 fictional entities run through the full pipeline) puts the empirical noise floor at ~ 0.04 for dense-gold cohorts, rising to ~ 0.20 for cohorts whose gold answer is short boilerplate (Section 6.7.7); scores must be read against that floor, not against zero.

3. Captures the citation residual. IKP Sec. 5.7 showed bibliometrics explain $\sim 35\%$ of recognition; the remaining residual decomposes into name uniqueness, named-artifact attachment, and subfield-density. NameRank measures this residual directly. For a working researcher in a dense-derivative-content subfield, the marginal effect of one widely-used open-source tool with clean name attribution exceeds the marginal effect of one well-cited paper. This is verified directly in our data (Section 6.4, Section 6.3).

3.3 What NameRank Does Not Measure

We are explicit: NameRank is not a quality, merit, or intrinsic-impact metric. It measures *the propagation of a specific name into the indexed corpora that frontier-model training pipelines scrape*. This correlates with academic impact through bibliometric mechanisms, with cultural reach through named-artifact attachment, and with both through derivative-content density (tutorials, blog posts, talks, podcast mentions). It does not measure the underlying work. The clearest case is C6 (tuixue.online): the artifact has ∼1M Chinese-language users—measurable, indisputable human reach—but registers NameRank 0.25 because the URL does not co-travel with its creator's name in indexable text. NameRank correctly registers low here, not because the artifact lacks impact, but because the metric measures something specific. We treat this as a feature: a clean instrument that reports what it measures, not what users wish it measured.

4 Methodology

Figure 2 summarizes the end-to-end NameRank pipeline at the level of detail this section unpacks: a fixed open-ended probe is sent to a 37-model panel for each entity, every (entity, model) response is scored by an independent LLM judge on multiplicative coverage×accuracy against a curated gold answer, embedding similarity is computed in parallel as a sanity check, and the per-entity NameRank is the unweighted panel mean of $\mathrm{cov} \cdot \mathrm{acc}$.

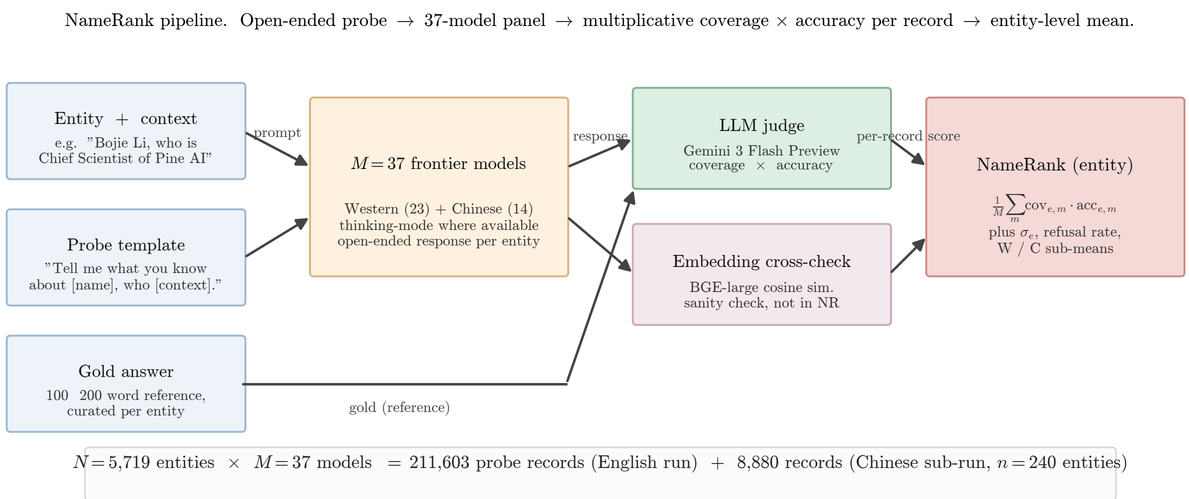

Figure 2. NameRank pipeline. The probe template is fixed with two slots (entity name; short role/affiliation/field context). The $M = 37$ models span Western (23) and Chinese (14) vendors with thinking-mode enabled where available. Each response is graded by an LLM judge against the gold answer on independent coverage and accuracy axes in $[0, 1]$; their product is the per-record contribution. The embedding cross-check uses BAAI/bge-large-en-v1.5 cosine similarity and is retained for diagnostics only—it is *not* part of the NameRank computation (Section 6.7). Total: 211,603 English-prompt records over 5,719 entities, plus an 8,880-record Chinese-prompt sub-run on a 240-entity subset.

4.1 Probe Design

One open-ended probe per entity. Closed-factoid probes have a known failure mode: a model that recognizes the entity through fact X but does not recall fact Y scores zero on a probe-of-Y, biasing the metric toward specific-fact-recall. Open-ended probes capture the holistic question "does the model have a retrievable representation of this entity?" The probe template is fixed (Section 3; full English and Chinese probe text plus the judge prompt are reproduced verbatim in Appendix B) with two slots: name (the canonical entity name) and context (a short disambiguating clause of role + affiliation + field, e.g., *"Bojie Li, who is Chief Scientist of Pine AI working on AI infrastructure"*; *"Wei Zhang, who is a professor of mathematics at MIT working on PDEs"*). The disambiguating context resolves name collisions at probe time without revealing the gold answer.

Why disambiguation does not contaminate scoring. The context names role/affiliation/field but not specific contributions, papers, dates, or named artifacts. A model that recognizes the entity will produce specific contributions in the response (which the judge scores against the gold); a model that does not recognize the entity will produce generic statements about the role/field (e.g., *"Bojie Li is a researcher in AI infrastructure"* with no specifics), which the judge correctly assigns low coverage because the rubric requires *specific* named facts (Section 4.3). The clearest empirical check is the bottom-tier cohort: the GPT-5 system-card-author probes include the disambiguating context "contributor listed in the GPT-5 system card," which is uniform across all 80 entities; nonetheless, 71% of probes return "unknown" rather than producing a generic in-context bio. The context provides anchoring but not score leakage.

4.2 Gold Answers

For each entity we construct a 100–200 word gold answer covering notable contributions, affiliations, and identifying facts. Source allocation by tier:

- **Wikipedia-covered entities** ($\sim$ 60% of cohort): Wikipedia first paragraph plus relevant key-contribution sentences, auto-extracted via the MediaWiki API and lightly edited.
- **OSS / GitHub artifacts** ($\sim$ 10%): README first paragraph + repository description + creator attribution.
- **Industry figures and Western mid-tier** ($\sim$ 10%): company about-pages, public role descriptions, news coverage.

- **Long-tail / Chinese-only / niche** ($\sim$ 20%): manual curation from Zhihu, personal sites, lab pages, competition archives, OpenAlex profiles.

Gold-answer quality matters less than one might expect because the judge's coverage score measures *which of the gold's named facts appear in the response*: if the gold omits a true contribution, both responses and judge are still evaluated against the same reference, so omission affects scale but not within-cohort ranking. One consequence of the source allocation is worth making explicit: for the $\sim$60% of entities whose gold derives from Wikipedia, coverage partly measures whether the model has memorized the same Wikipedia text that seeds the gold—a source/training overlap that is intended (Wikipedia presence *is* part of corpus presence) but means the dense-gold cohorts are scored against a reference their recognition correlates with by construction. The Wikipedia-confound check (Appendix N) shows this overlap does not reduce NameRank to a has-a-page flag: a binary Wikipedia indicator explains only $\sim$ 2% of long-tail-researcher variance against 40% for h-index. We audit a stratified 5% sample ($n \approx 286$) for factual errors against primary sources; the observed correction rate is $\sim$ 3% (typically year-of-affiliation drift). The auditing protocol parallels the Wikidata-audit pipeline of Li [2026].

4.3 The Judge

We use Gemini 3 Flash Preview (direct Google API, not via OpenRouter, to minimize routing variance) as the judge, with temperature 0 and low reasoning budget. The judge prompt presents the gold and the response, and elicits two independent scores in $[0,1]$:

- **Coverage** (0.0–1.0): how much of the gold's *substantive* content appears in the response. "Substantive" = named artifacts, named affiliations, named contributions, identifying facts (year, location, role title). Vague statements like *"works in AI"* are not substantive content.
- **Accuracy** (0.0–1.0): are the factual claims in the response correct, per the gold. A "factual claim" is a specific named entity, year, role, or relationship. Hallucinated bios with high coverage but wrong specifics score low on accuracy.

The judge prompt includes the explicit hallucination instruction: *"Hallucinated bios (high coverage, low accuracy) MUST score LOW on accuracy."* Refusals (*"I don't know", "unknown"*) score 0.0 on both axes. Correct extra facts that are not in the gold are not penalized (the judge is told this explicitly); vague statements that could apply to many entities are not credited.

Judge family-bias check. We use Gemini 3 Flash Preview as the single judge for the full evaluation. To confirm this does not bias the findings, we re-judge a 615-record stratified sample with two further judges from different families (Claude Opus 4.6, GPT-5.5). The three agree at pairwise Pearson 0.87–0.93 and preserve the cohort ladder; a capability-controlled own-family lift exists but is small (Gemini judging Gemini-family responses $+0.062$, larger than a naive within-entity estimate suggests), and it shifts per-vendor score tables without moving the cross-entity comparisons the paper draws. Full analysis in Section 6.7.6 and Appendix L.

4.4 Embedding Cross-Check

We additionally compute the cosine similarity between the response and the gold answer using BAAI/bge-large-en-v1.5 [Xiao et al., 2024], stored as a sanity check at the per-(entity, model) record level. Embedding similarity provides a fast, judge-free signal whose disagreements with the judge are diagnostically informative (Section 6.7). Embedding similarity is *not* part of the NameRank computation itself—we explicitly use multiplicative coverage$\times$accuracy with the LLM judge—because, as we show, embedding similarity systematically credits fluent hallucinations as recognized.

4.5 Aggregation

NameRank is the unweighted mean of $\mathrm{cov} \cdot \mathrm{acc}$ across the 37-model panel. We also retain the within-entity standard deviation σ_e (cross-model disagreement), the refusal rate (fraction of panel responding "unknown"), the Western/Chinese sub-means (Section 6.5), and the raw response text. The score is in $[0,1]$ by construction; the empirical range across our 5,719 entities is $[0.000, 1.000]$.

Why averaging across models is correct. A single model produces score noise from prompt-sensitivity, refusal-policy idiosyncrasies, and training-data accidents. The panel mean is a stable estimate of corpus presence integrated over the current frontier model fleet. Within-entity variance retains the corpus-pocket-specific signal (an entity recognized by Gemini but not GPT-5 is empirically interesting; the panel mean smooths this; the σ_e surfaces it).

5 Experimental Setup

5.1 Model Panel

$M = 37$ frontier and near-frontier models accessed via OpenRouter (probed models) and direct Google API (judge). Thinking mode enabled wherever a thinking variant exists, departing from IKP's parameter-estimation default of non-thinking variants—for NameRank, thinking-mode is the operational mode in which most user queries are answered, so the panel reflects deployment reality. The panel is balanced across vendors:

Western (n=23)	**Chinese (n=14)**
GPT-5.3, GPT-5.4, GPT-5.5-think, GPT-OSS-20b-think	DeepSeek-V3.2-think, V4-flash-think, V4-pro-think
Claude Opus 4.6-think, Sonnet 4.6-think	Qwen3-8b-think, 32b-think, 235b-a22b-think, 3.5-397b-a17b-think
Gemini 2.5-pro-think, 3-flash-think, 3.1-pro	GLM-4-32b, 4.7-think, 5.1-think
Llama-3.1-8b, 3.2-1b, 3.3-70b, 4-Maverick	Kimi-K2, K2.6-think
Mistral-Large, Medium 3.1, Small 24b, Ministral 3b	Ernie-4.5-300b-a47b
Gemma-3-4b, 3-12b, 4-31b	MiniMax-M2.7-think
Grok-4, Grok-4.20-think, Phi-4	

The Western/Chinese partition is used for East-West variance analysis (Section 6.5); it does not enter the headline NameRank computation, which is the unweighted panel mean.

5.2 Entity Cohort

$N = 5{,}719$ entities across 54 cohorts. Top-level structure:

- **Credentialed individuals** ($n \approx 801$). IMO gold ($n = 197$), IOI gold (68), ICPC World Finalist gold (18), Putnam top-25 (35), CMO China gold (131), NOI China gold (29), CPhO China first prize (50), Rhodes Scholar (36), MSRA PhD Fellowship (237).
- **Working academics** ($n \approx 1{,}628$). CS faculty (698, sampled from CS Rankings + OpenAlex), OpenAlex long-tail researchers (771, citations 500–30,000), long-tail IKP researchers (159, drawn from the IKP Sec. 5.7 345-researcher subset, minus those already covered in cs_faculty).
- **Industry founders, VCs, AI-policy figures** ($n \approx 248$). AI startups/companies (102), mid-tier founders (48), VCs (31), YC sample (25), AI policy / government working level (42).
- **Named artifacts** ($n \approx 1{,}332$). Foundation models (107), benchmarks (47), datasets (45), named methods (73), OSS projects (184), research papers (298), long-tail papers (499, 50–500 citations), programming languages (29), databases / data systems (30), AI hardware (20).
- **Mid-tier cross-profession** ($n \approx 1{,}461$). Comedians (46), chefs (47), filmmakers (38), journalists (84), politicians (103), historical figures (105), medical figures (44), religious figures (30), online courses (40), podcasts (57), books (130), writers (194), musicians (64), athletes (134), actors (123), artists (45), architects (36), products (88), OSS maintainers (51), activists (2, retained as a placeholder for future expansion but excluded from cohort-level claims).
- **Diagnostic entities** ($n = 37$, `reference_pilot` cohort, hand-curated higher-detail gold answers). *Public figures*: Sam Altman, Geoffrey Hinton, Yann LeCun, Demis Hassabis, Dario Amodei, Mira Murati, Greg Brockman, Aravind Srinivas, Aman Sanger, Andrej Karpathy, Tri Dao, Simon Willison, Lilian Weng, Chris Olah, Harrison Chase, Logan Kilpatrick. *Less-prominent*: Wojciech Zaremba, Trevor Blackwell, Vicki Cheung, Pamela Vagata, Aaditya Singh, Adam Fry, AJ Ostrow, Alex Wei, Andrea Vallone, Aleksander Madry, Bojie Li, Jiayi Weng, Suresh Kumar. *Artifacts*: Tianshou, tuixue.online, IKP, FlashAttention, ReAct, LangChain, vLLM, nanoGPT.
- **Special diagnostic + auxiliary cohorts** ($n \approx 212$). GPT-5 system card authors (80), DeepSeek-V3 paper authors (69), conferences (30), awards (15), industry products (9), public-utility websites/services (9).

The cohorts sum to 5,719 entities. Full cohort sizes and inclusion criteria are in Appendix A.

5.3 Cross-Language Sub-Run

A subset of 240 entities was re-probed in Chinese (probe template translated, gold answer left in English, judge prompt left in English—the judge handles cross-language grading). Cohort selection: all reference_pilot (37), all NOI China gold (29), all DeepSeek-V3 paper authors (69), 30 random samples each from CMO, CPhO, MSRA Fellowship, plus 5 each from mid-tier-writer/filmmaker/politician as Western controls. Total Chinese-prompt records: $240 \times 37 = 8{,}880$.

5.4 Execution

The pilot runner is a 24- to 96-way parallel Python process using OpenRouter for probed models, direct Google API for the judge, and the local BGE-large model for embeddings. The full English run completed in $\sim$10 hours of wall-clock time at 96-way parallelism after a checkpoint resume from an earlier 24-way run (no records lost). The Chinese run completed in $\sim$30 minutes. Cost (probed-model API calls + judge calls, computed at per-vendor rates): $\sim$\$2,500 end-to-end including iteration costs.

5.5 Reproducibility

We release: (i) the 5,719-entity probe specification with disambiguating contexts, (ii) the 5,719 gold answers, (iii) the 211,603 raw response records with judge scores and embedding similarities, (iv) the analysis pipeline, (v) the Chinese sub-run results. Sources: `https://github.com/19PINE-AI/namerank`. Companion website: `https://01.me/research/namerank`.

6 Results

6.1 Cross-domain Headline

Figure 1 shows the per-cohort NameRank distribution. The instrument resolves into three zones cleanly visible on the same axis: *silent* (≤ 0.1, dominated by paper-published-this-month entities such as the GPT-5 system-card author cohort), *discriminative* (0.3–0.6, working academics and credentialed individuals), and *universal* (≥ 0.7, named artifacts and high-profile mid-tier figures).

The cross-domain placement is the headline result. On a single axis (Table 1):

Table 1. Selected cross-domain entities on a single NameRank axis. No prior instrument places academic researchers, industry founders, OSS authors, public-utility websites, and a sub-top-50 CTO on the same scale.

Entity	NameRank	Cohort/role
Yuval Noah Harari	1.000	mid_tier_writer (popular author)
Sam Altman	0.95	reference_pilot (CEO, OpenAI)
Geoffrey Hinton	0.93	reference_pilot (deep-learning pioneer)
Aravind Srinivas	0.66	reference_pilot (Perplexity CEO)
Andrej Karpathy	0.61	reference_pilot (Eureka Labs founder)
Tri Dao	0.53	reference_pilot (FlashAttention; Princeton faculty)
Tianshou	0.42	reference_pilot (RL library)
Jiayi Weng	0.33	reference_pilot (Tianshou author; OpenAI infra)
Suresh Kumar (Walmart CTO)	0.27	reference_pilot (sub-top-50 CTO)
tuixue.online	0.25	reference_pilot (Chinese visa-monitoring site)
Bojie Li	0.09	reference_pilot (this paper's author)
GPT-5 system-card-author mean	0.042	gpt5_system_card_author cohort

No prior instrument supports this placement. Citation count does not measure Altman, Srinivas, or Suresh Kumar; *h*-index does not measure the creator of nanoGPT; GitHub stars do not measure Yuval Harari; book sales do not measure Tri Dao. NameRank does measure them all, on the same scale.

6.2 The Credential Treadmill Thesis

Nine prestigious credentials, ranked by mean NameRank with the long-tail-OpenAlex-researcher cohort as baseline (Table 2):

Table 2. Credential ladder. Long-tail OpenAlex researchers ($n = 771$, citations 500–30,000) supply the baseline. Most prestigious-credential cohorts sit *at or below* the working-researcher baseline.

Credential	n	Mean NameRank	vs. baseline	vs. Yuval Harari
ICPC World Finalist gold	18	0.610	+0.146	−0.390
Putnam top-25 fellow	35	0.608	+0.144	−0.392
Long-tail OpenAlex (baseline)	**771**	**0.464**	—	−0.536
NOI China gold	29	0.368	−0.096	−0.632
IMO gold (2005–2015)	**197**	**0.362**	**−0.102**	−0.638
MSRA PhD Fellowship	237	0.343	−0.121	−0.657
IOI gold	68	0.341	−0.123	−0.659
Rhodes Scholarship	36	0.315	−0.149	−0.685
CMO China gold	131	0.302	−0.162	−0.698
CPhO China first prize	50	0.182	−0.282	−0.818
DeepSeek-V3 paper author	69	0.178	−0.286	−0.822
GPT-5 system card author	80	0.042	−0.422	−0.958

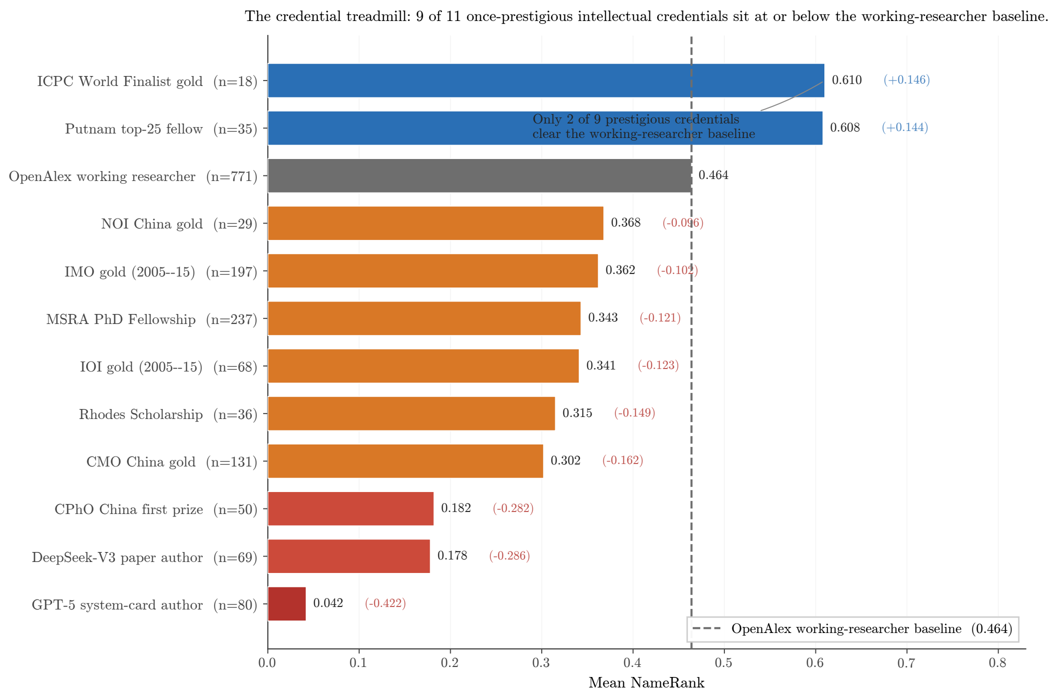

Figure 3. The credential treadmill (graphical form of Table 2). Bars are cohort mean NameRank; the dashed grey line is the long-tail-OpenAlex working-researcher baseline (0.464). Red bars: silent or near-silent cohorts. Orange: at or just below the baseline. Blue: above the baseline. Of nine prestigious-credential cohorts, only two (ICPC, Putnam) clear the working-researcher baseline.

The pattern is robust (Figure 3, Table 2). Seven of nine credentialed cohorts sit *below* the long-tail-OpenAlex working-researcher baseline. IMO gold ($n = 197$, the canonical intellectual credential of the past four decades) has mean NameRank 0.362, lower than the average citation-tracked researcher (0.464). Rhodes Scholars ($n = 36$) sit at 0.315, MSRA PhD Fellowship holders ($n = 237$) at 0.343.

Baseline-selection caveat. The two cohorts are not selected symmetrically, and the asymmetry runs *toward* the finding. The OpenAlex baseline is conditioned on the outcome correlate—it includes only researchers who already cleared a 500-citation floor—whereas the credential cohorts are unconditioned on career outcome and include holders who left research or never published. A citation-floored comparison group will mechanically outscore an unfiltered group on a citation-correlated metric, so part of the raw −0.102 IMO-vs-baseline gap reflects this selection floor rather than credential decay per se. Two facts keep the thesis intact under the conservative reading. First, the credential cohorts are not merely *below* the citation-selected baseline; the lowest (CPhO 0.182, DeepSeek-V3 authors 0.178) sit near the short-gold noise floor (Section 6.7.7), an absolute statement that does not depend on the baseline. Second, conditioning on career track sharpens rather than reverses the effect: IMO golds on US-tracked academic/industry careers—the subset most comparable to the OpenAlex cohort—score ∼0.50, and the gap to 0.30 for non-Anglophone-tracked golds is the mechanism (below), not a baseline artifact. A fully outcome-matched baseline (golds restricted to those who became citation-tracked researchers) is the cleaner future comparison; we report the unconditioned cohorts here because credential-as-signal is the population an institution actually selects on.

Why ICPC and Putnam are exceptions. Both feed into US-tracked academic/industry careers with high English-corpus density. ICPC World Finalists ($n = 18$) become competitive programmers at FAANG companies and quant funds, with English-language profiles propagating well; Putnam top-25 fellows ($n = 35$) typically pursue US PhD programs in mathematics. These are not credential exceptions; they are proxies for selection into high-English-text-density career tracks.

Why IMO/CMO are not exceptions despite identical intellectual selection. IMO and CMO golds disperse globally, often becoming non-English-speaking researchers, mathematicians or engineers whose post-medal career produces text in languages other than English (German, Russian, Chinese, Korean, Japanese, Romanian, Farsi, etc.). The medal-and-name pair never co-occurs at high density in English training corpora. Conditioning on country reveals the gradient: IMO golds from US-tracked careers score ∼0.50; from non-Anglophone careers ∼0.30.

MSRA Fellowship temporal pattern. Within MSRA fellows, 5-year-cohort means rise from 0.312 (2005–2009) to 0.337 (2010–2014) to 0.382 (2015–2019) and dip to 0.343 (2020–2024)—consistent with the more recent fellows having had less time for post-fellowship career production to propagate. Pearson(year, NameRank) within IMO 2005–2015 is -0.042, indistinguishable from zero: the credential signal is not strengthening or decaying with time within the 10-year window.

Generational claim. Olympic-style credential systems were optimized for a pre-LLM signaling regime in which credentials translated into reputation through human social networks. In the LLM-mediated regime, the signal channel has shifted: only credentials that translate into *named, citable productions*—papers with the holder's name attached, OSS projects, blog posts, podcasts—propagate. The credential itself, absent subsequent production, does not.

6.3 The Named-Artifact-Amplification Mechanism

6.3.1 Artifact > creator inversion

Table 3 shows 11 verified (creator, artifact) pairs:

Table 3. Artifact > creator inversion. Columns: NR_c = creator NameRank, NR_a = artifact NameRank, $C \to A$ = fraction of model responses about the creator that mention the artifact, $A \to C$ = reverse. Boldface: the artifact's NameRank exceeds its creator's.

Creator	NR_c	Artifact	NR_a	$\Delta(a-c)$	$C \to A$	$A \to C$
Simon Willison	0.594	**Datasette**	**0.899**	**+0.305**	0.54	0.89
Dario Amodei	0.584	**Anthropic**	**0.811**	**+0.228**	1.00	0.41
Andrej Karpathy	0.614	**nanoGPT**	**0.707**	**+0.093**	0.05	0.76
Jiayi Weng	0.331	**Tianshou**	**0.420**	**+0.089**	0.25	0.00
Tri Dao	0.532	**FlashAttention**	**0.607**	**+0.075**	0.74	0.54
Lilian Weng	0.590	**lilianweng.github.io**	**0.614**	**+0.024**	0.97	0.97
Harrison Chase	0.479	**LangChain**	**0.502**	**+0.024**	1.00	0.22
Aman Sanger	0.289	**Cursor**	**0.305**	**+0.016**	1.00	0.00
Demis Hassabis	0.664	Google DeepMind	0.490	−0.174	1.00	0.62
Mira Murati	0.442	Thinking Machines Lab	0.222	−0.220	0.97	1.00
Aravind Srinivas	0.663	Perplexity	0.193	−0.470	1.00	0.33

In 8 of 11 pairs, the artifact's NameRank exceeds the creator's (Table 3). The three exceptions are senior leaders of large named-product organizations (Hassabis/DeepMind, Murati/Thinking Machines, Srinivas/Perplexity), where the leader has substantial independent corpus presence from earlier credentials and the new organization is too young to have built equivalent name-attribution mass. For independent creators of single artifacts (Datasette, nanoGPT, Tianshou, LangChain, Cursor, FlashAttention, lilianweng.github.io), *the artifact uniformly exceeds the creator*.

6.3.2 Conditional-probe asymmetry

The same table's last two columns quantify the direction of attribution flow. $C \to A$ is the fraction of model responses to "tell me about [creator]" that name the artifact. $A \to C$ is the reverse. Per-pair, per-model mention breakdowns are tabulated in Appendix D. Two regimes are visible:

- **Famous artifact, anonymous creator** ($C \to A$ high, $A \to C$ low): Aman Sanger / Cursor (+1.000 asymmetry), Harrison Chase / LangChain (+0.78), Aravind Srinivas / Perplexity (+0.67), Dario Amodei / Anthropic (+0.59). When probed about the creator, models reliably name the artifact; when probed about the artifact, models often fail to name the creator.
- **Paired in corpus** ($C \to A$ and $A \to C$ both high): Lilian Weng / lilianweng.github.io (0.97/0.97), Mira Murati / Thinking Machines Lab (0.97/1.00), Tri Dao / FlashAttention (0.74/0.54). The academic-norm pattern: creator and artifact co-occur reliably in both directions.
- **Inverse asymmetry**: Andrej Karpathy / nanoGPT ($C \to A = 0.05$, $A \to C = 0.76$). Karpathy is famous for too many things (Tesla, OpenAI, Stanford lectures); a probe about him rarely surfaces nanoGPT specifically, while probes about nanoGPT reliably surface Karpathy as the originator. This is structurally distinct from the famous-artifact-anonymous-creator regime.

6.3.3 Per-response mediation

On the Jiayi Weng probe specifically, mentioning Tianshou directly mediates the score:

Model class	n	Mean score
Models that mention "Tianshou"	7	**0.71**
Models that do not mention "Tianshou"	21	**0.35**

The seven mentioning models are exactly the frontier-reasoning subset: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. This is the limit of the observational comparison: because the mentioning models *are* the most capable ones, the +0.36 gap conflates "names the artifact" with "is a stronger model that both recognizes Jiayi and recalls his tool," and cannot by itself establish that the artifact mention is what carries the score. We read this table as a suggestive per-response signature only; the stronger evidence rests on the context-injection experiment below (Section 6.3.4), which holds the model fixed and manipulates the artifact mention directly.

6.3.4 Context-injection experiment

The mediation evidence above is observational. To manipulate the artifact mention directly, we re-probed all 11 verified (creator, artifact) pairs under a controlled intervention: arm A uses the creator's standard disambiguating context; arm B appends the artifact name to it (e.g. adding "creator of Tianshou"), holding everything else fixed. Injecting the artifact name raises the creator's NameRank by a mean +0.058 (median +0.029, positive in 8 of 11 pairs). The effect is largest exactly where the observational analysis predicts: Jiayi Weng +0.288 (paired $t = 5.8$), the manipulated counterpart of his +0.369 observational per-response lift; Harrison Chase / LangChain +0.151; Tri Dao / FlashAttention +0.094. It is null or negative for the three senior leaders whose baseline recognition is already high (Amodei −0.047, Srinivas −0.022, Murati −0.014): the mechanism cannot push a model above what it already knows. Consistently, the lift runs through retrieval-deficient models—by tier it is +0.008 (frontier-reasoning), +0.046 (frontier-chat), +0.106 (mid-weight), +0.095 (small). Naming the artifact is load-bearing precisely for the models that have not independently stored the creator.

A coverage-echo confound, and what the experiment does and does not establish. Because the injected clause names the artifact, and the gold answer also names it, a model can raise its coverage score simply by echoing the injected token—"Jiayi Weng created Tianshou"—without any additional stored knowledge. Part of arm B's lift is therefore mechanical re-scoring of a fact handed to the model in the prompt, not retrieval amplification, and the released aggregates (per-(entity, model) products only) do not let us separate the two post hoc. This is the same scoring-leakage channel we otherwise close for the standard probe by withholding contributions from the context (Section 4.3); here we deliberately open it. Two features of the result limit the confound but do not eliminate it: the lift is concentrated in low-baseline cases rather than spread uniformly (pure echo would add coverage to known entities too, yet the saturated leaders show none), and the tier gradient tracks recognition deficit. The clean test—which we register for the next run rather than claim here—is to re-judge arm B against an *artifact-redacted* gold that credits only the facts *not* present in the injected clause; the residual lift on those facts is the leakage-free amplification estimate. Pending that, we read the experiment as *consistent with* a causal named-artifact-amplification mechanism, with an upper-bounded effect size, rather than as a clean causal identification.

6.4 External Validity

6.4.1 h-index dominates raw citations

On the long-tail-OpenAlex-researcher cohort ($n = 771$, citations 500–30,000, with h-index from OpenAlex):

Predictor	R^2 (within cohort)
log(citations)	0.137
log(h-index)	**0.398**
log(citations) + log(h-index)	0.398

Joint regression coefficients: log(h-index) +0.138, log(citations) −0.002. **Citations add zero marginal predictive power beyond h-index.**

Out-of-sample validation (repeated 10-fold cross-validation, 20 repeats / 200 folds on the same cohort): held-out $R^2 = 0.38 \pm 0.09$ for log(h-index) and 0.12 ± 0.07 for log(citations); the joint model matches log(h-index) alone (0.38 ± 0.09). The relationship is stable out of sample, and citations carry no held-out signal. (We report cross-validated R^2 rather than a single train/test split, whose estimate swings ∼0.15 with the seed at this sample size.) Full regression and per-decile breakdown are in Appendix F.

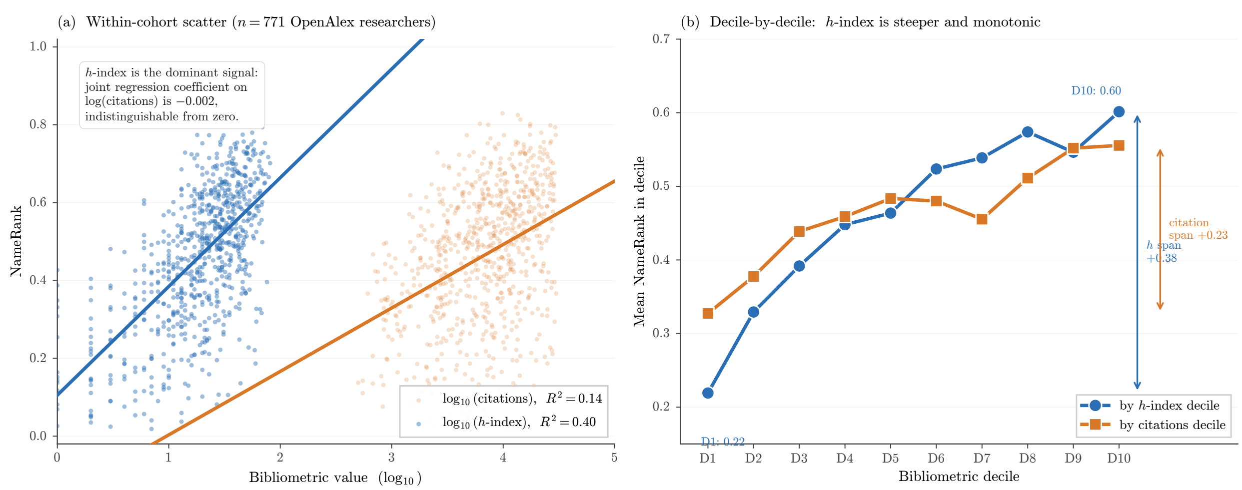

Figure 4. External validity: h-index dominates raw citations as a NameRank predictor. **(a)** Scatter of NameRank against $\log_{10}$(h-index) (blue) and $\log_{10}$(citations) (red) for the $n = 771$ OpenAlex long-tail-researcher cohort. The h-index regression has a steeper slope and explains $R^2 = 0.40$ of variance; the citation regression explains only $R^2 = 0.14$. **(b)** Mean NameRank by decile of each bibliometric: the h-index ladder is monotonic from 0.22 (D1) to 0.60 (D10), while citations show a much flatter 0.33-to-0.56 slope. In joint regression, citations contribute zero marginal R^2 beyond h-index.

Mechanistic interpretation. h-index measures the *number of distinct works each carrying non-trivial citation mass*—a sustained pattern of citable artifacts that recur, under the author's name, across the indexed corpus. We tested the natural alternative mechanism—that h-index wins because it implicitly *discounts per-paper name attribution* (the hyperauthorship problem [Cronin, 2001], in which a viral thousand-author paper inflates raw citations while diluting individual name attribution)—and it does not hold (Figure 5, Appendix K). On the same $n = 771$ cohort, explicitly attribution-weighted citation counts barely improve on raw citations (fractional-citation $R^2 = 0.154$ vs. raw-citation 0.137; first/last-author 0.268) and stay well below h-index (0.398); the mean-authors-per-paper regressor carries essentially zero signal ($R^2 = 0.0004$), and adding fractional citations to h-index lifts R^2 by only 0.003. What *does* track h-index is the raw count of distinct works (n_{works} alone gives $R^2 = 0.337$). NameRank is therefore a *name-recurrence* metric: it rewards how many distinct citable works repeat the author's name, not how cleanly attribution is divided within any single paper. Citation volume remains the wrong feature; the refinement is that the right one is recurrence count, not attribution-per-paper weighting.

IKP Sec. 5.7 replication, refined. Li [2026] reports that bibliometrics explain ∼35% of recognition variance, with citations as the bibliometric. With h-index instead, we explain 40%. The original ∼35% figure was an average across these features; the right decomposition is that the h-index channel carries the bibliometric signal and raw-citation channel carries nearly none of it.

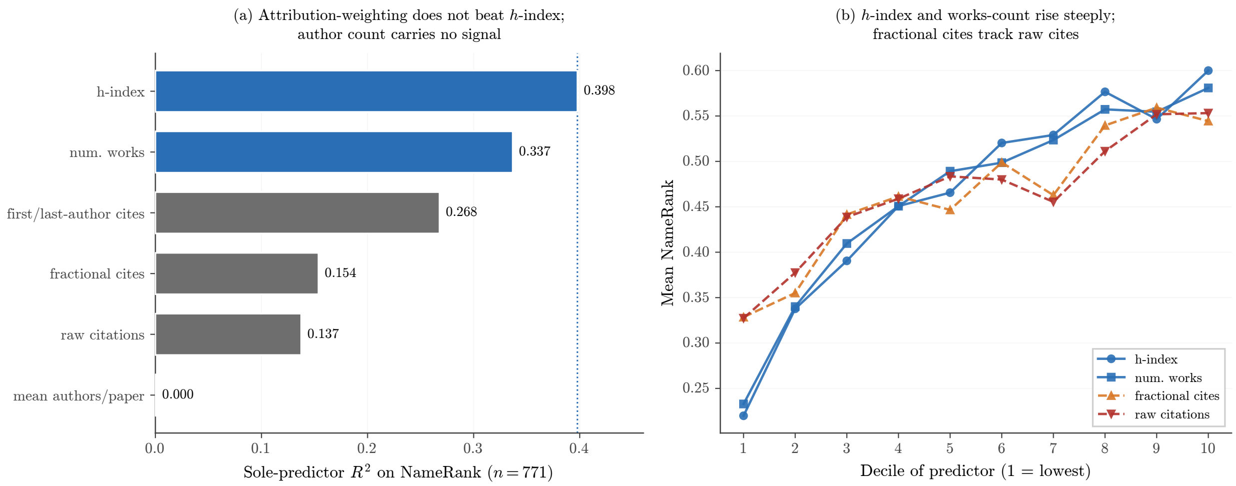

Figure 5. Bibliometric predictor decomposition ($n = 771$ OpenAlex long-tail cohort). **(a)** Sole-predictor R^2 on NameRank. If h-index dominated because it discounts per-paper name attribution, the attribution-weighted measures (fractional and first/last-author citations) would match it; instead they barely beat raw citations, and mean-authors-per-paper carries no signal. The count of distinct works (n_{works}) tracks h-index most closely. **(b)** Mean NameRank by predictor decile: h-index and n_{works} rise steeply and monotonically, while fractional citations track raw citations on a flatter slope. The mechanism is name-recurrence across distinct works, not attribution-per-paper weighting (Appendix K).

6.4.2 Institution and country gradient

Within the CS faculty cohort ($n = 698$ across 320 distinct institutions), affiliation is a substantial correlate of NameRank, but its variance share must be read with care. A naive categorical regression with one dummy per institution returns $R^2 = 0.544$; this is inflated by the category count—319 free parameters fit to 698 observations (∼2.2 faculty per institution), so that the expected R^2 under a pure-noise null is already 0.458. The penalized estimates are far lower and mutually consistent: adjusted $R^2 = 0.159$, $\omega^2 = 0.159$, and the one-way random-effects intraclass correlation is 0.161. We therefore report that **institution explains** ∼16% **of within-cohort NameRank variance, not** 54%. What survives cleanly is the large-cell contrast (Table 4): Stanford-affiliated faculty ($n = 19$) average 0.537 against Tsinghua's 0.258 ($n = 4$), a ∼2× gap, and the country-level aggregation below—which uses far fewer levels and is not subject to the same inflation (country raw $R^2 = 0.144$, adjusted 0.105)—shows the gradient at population scale.

Table 4. CS faculty NameRank by institution (15 selected institutions with $n \geq 4$). Stanford-affiliated faculty score 2× Tsinghua-affiliated. The Tsinghua/Peking samples are small ($n = 4$ each); the gradient holds across larger samples (CMU, Cornell, Michigan, Princeton, etc.).

Institution	n	Mean	Min	Max
Stanford University	19	**0.537**	0.14	0.62
University of Washington	26	0.529	0.13	0.68
Cornell University	33	0.484	0.16	0.67
Carnegie Mellon University	65	0.484	0.09	0.66
Princeton University	25	0.443	0.18	0.61
HKUST	5	0.462	0.24	0.58
Johns Hopkins University	4	0.453	0.19	0.67
TU Delft	5	0.445	0.30	0.58
University of Michigan	31	0.429	0.16	0.67
Universidade de Lisboa	5	0.427	0.26	0.52
Georgia Institute of Technology	6	0.462	0.23	0.65
Peking University	4	0.290	0.22	0.34
Tsinghua University	4	**0.258**	0.15	0.35
USTC	4	0.310	0.11	0.45
Monash University	6	0.386	0.27	0.65

Country-level gradient. Aggregating institutions by country (Table 5, Figure 6):

Table 5. CS faculty NameRank by country of institutional affiliation. Countries with ≥ 3 sampled faculty; `Other/Unknown` (n=236) consists of CS-Rankings affiliations not matched by the keyword-prefix country lookup and is omitted from the gradient discussion. Only USA ($n = 302$), China (30), UK (16), Australia/Hong Kong (11), Brazil/India (10) reach $n \geq 10$; the remaining rows are point estimates with wide intervals and should not be finely ranked against one another.

Country	n	Mean	Median	frac ≥ 0.5
Israel	7	0.506	0.528	57%
Singapore	7	0.497	0.480	43%
Sweden	5	0.492	0.490	40%
Germany	9	0.490	0.488	44%
Switzerland	4	0.489	0.467	50%
Canada	9	0.478	0.463	22%
USA	302	0.460	0.488	46%
Netherlands	9	0.447	0.432	33%
UK	16	0.438	0.440	38%
Hong Kong	11	0.431	0.452	36%
Portugal	5	0.427	0.442	20%
Brazil	10	0.388	0.409	10%
Australia	11	0.341	0.291	18%
France	3	0.312	0.341	0%
China	30	**0.311**	0.285	7%
Spain	8	0.303	0.288	13%
India	10	0.292	0.279	0%

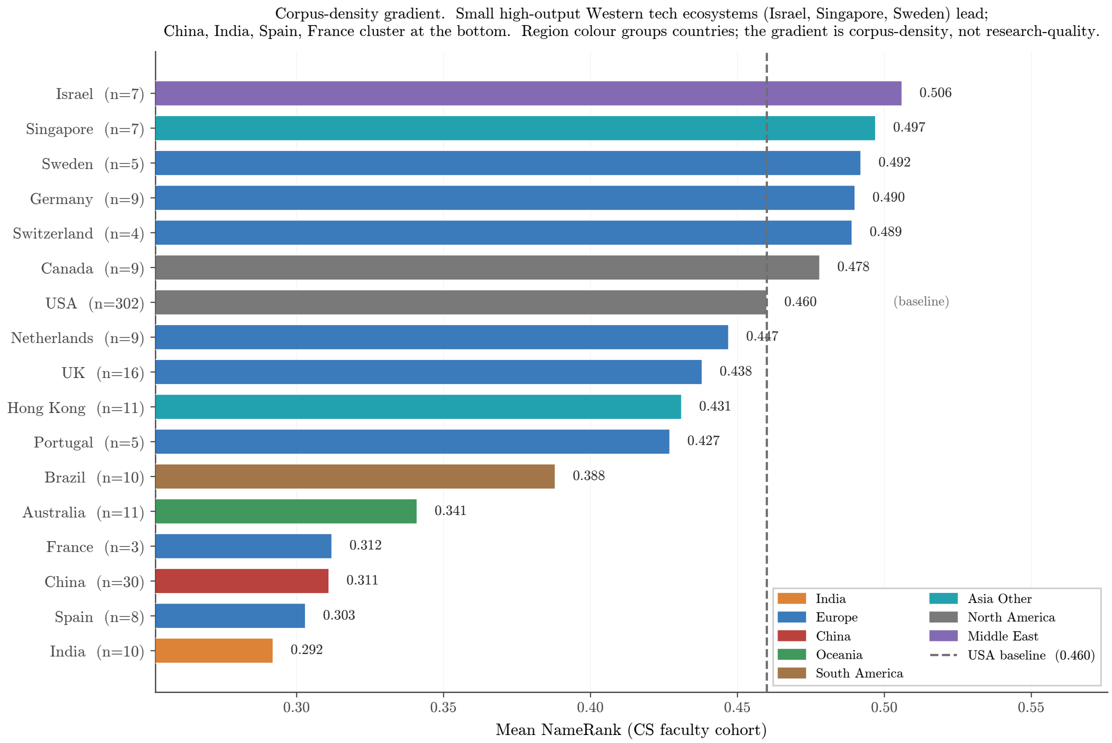

Figure 6. Country gradient in CS faculty NameRank. Small high-output Western tech ecosystems (Israel, Singapore, Sweden; $n = 5$–7 each) score above the USA baseline (dashed) in point estimate but with intervals that overlap it; the firmly-separated contrast is USA ($n = 302$) vs. China ($n = 30$) and India ($n = 10$), whose 95% CIs do not overlap the USA's. Chinese, Indian, French, and Spanish CS faculty cluster at the bottom. The gradient is corpus-density, not research-quality: Tsinghua, Peking, USTC and similar institutions have global research reputations comparable to upper US institutions but produce far less English-language web text per affiliation. Countries with $n < 10$ (Israel, Singapore, Sweden, Germany, Switzerland, Canada, Netherlands, Portugal, France, Spain) are point estimates only; rank them with caution.

The gradient is corpus-density, not research-quality. Tsinghua is China's top CS school; the gap to Stanford is not a quality gap. It is a measure of English-language web-text production volume per affiliation. The firm, well-sampled part of the gradient is the gap from the USA ($n = 302$, mean 0.460, 95% CI $[0.443, 0.476]$) down to China ($n = 30$, 0.311, $[0.271, 0.350]$) and India ($n = 10$, 0.292, $[0.234, 0.351]$), whose confidence intervals do not overlap the USA's. The apparent leaders—small high-output Western tech ecosystems (Israel $n = 7$, Singapore $n = 7$, Sweden $n = 5$)—sit above the USA in point estimate but with wide intervals that overlap it, so we read them as suggestive rather than established; the ranked country ladder should be trusted only for the countries with $n \geq 10$. The substantive claim—that some Chinese institutions (Tsinghua, Peking, USTC) carry global research reputations on par with the upper US cluster yet score at the bottom—rests on the well-sampled China cell, not on the small-n tails.

The Tsinghua bind. A young Chinese CS faculty member at Tsinghua produces papers, lab pages, and conference talks that are equally rigorous as their CMU counterpart but propagate primarily through Chinese-language channels (Zhihu, WeChat technical articles, Chinese-language tutorials) plus their English papers. The CMU counterpart's same output additionally propagates through Twitter/X, Substack, English-language podcast appearances, US-news coverage of the institution, and a denser tutorial ecosystem. The CMU researcher's name-attribution mass per unit of research output is structurally higher. NameRank measures the result.

6.5 East/West Model Variance and Cross-Language Sub-Run

6.5.1 English-prompt East-West signal

Partitioning the 37-model panel into 23 Western and 14 Chinese models, the cohort-level median Chinese-minus-Western delta is *positive across nearly every cohort*, ranging from +0.02 to +0.10. The Chinese-minus-Western lift does not reflect Chinese models preferentially recognizing Chinese-content entities; rather, Chinese models commit globally to confident-but-accurate-enough descriptions where Western models more readily refuse on unfamiliar names. Cohorts with the largest positive C–W medians: ai_hardware (+0.096), mid_tier_architect (+0.067), research_paper (+0.062), ai_startup_or_company (+0.062), mid_tier_gov_ai_policy (+0.062).

Individual cases reveal more structure. Strongest Chinese-content lift: Wei Dongyi (Chinese math figure) C = 0.80 vs. W = 0.53 (+0.27); Zhang Fang (NOI China gold) C = 0.45 vs. W = 0.20 (+0.25). Strongest Western lift: Saad Albawi (Arab-named Western researcher) W = 0.59 vs. C = 0.16 (−0.42, Chinese models refuse on the name); Qingzi Vera Liao (Chinese-name CS faculty at Western institution) W = 0.50 vs. C = 0.15 (−0.35).

Interpretation. The Chinese-content lift is real but small (median +0.03–0.10 at cohort level), while the willingness-to-commit gap is large: the high-refusal cluster (gpt-oss-20b-think at 48%, grok-4.20-think at 46%, mistral-small-24b at 46%, qwen3-235b-a22b-think at 42%) refuses heavily and contains both Western and Chinese models; the zero-refusal cluster (gemma-3-4b, phi-4, ernie-4.5-300b-a47b at 0%, llama-3.2-1b/ministral-3b/llama-4-maverick at $\leq 0.1\%$) commits regardless and also spans both classes. The dominant signal is *threshold-of-confidence* asymmetry between training regimes (RL-tuned-against-hallucination vs. fluent-completion-by-default), not corpus-coverage asymmetry between Western and Chinese training data.

6.5.2 Chinese-prompt sub-run

We re-probed 240 entities with the Chinese-translated probe template (gold answer left in English; judge handles cross-language). The headline finding is the *prompt-language × entity-name interaction* (Table 6):

Table 6. Cross-language NameRank: $\Delta(\mathrm{NR}_{zh} - \mathrm{NR}_{en})$ by cohort. Chinese-prompt lifts Chinese-name entities and slightly suppresses English-coded entities.

Cohort	n	NR_{en}	NR_{zh}	Δ
deepseek_v3_author	69	0.178	**0.247**	**+0.069**
mid_tier_filmmaker	5	0.765	0.802	+0.037
noi_china_gold	29	0.368	0.392	+0.025
cmo_china_gold	30	0.301	0.315	+0.013
cpho_china_first_prize	30	0.184	0.194	+0.010
mid_tier_politician	5	0.361	0.364	+0.003
mid_tier_writer	5	0.650	0.631	−0.020
msra_phd_fellowship	30	0.339	0.319	−0.021
reference_pilot	37	0.452	0.422	**−0.030**

Per-entity extremes. Largest Chinese-prompt lifts: Wang Haoyu (+0.17), Yao Xuanrong (+0.16), Wang Zhongjian (+0.16), Zijia Zhu (DeepSeek author, +0.19). Largest Chinese-prompt suppressions: FlashAttention (−0.14), Aleksander Madry (−0.12), Du Zhuofan (−0.15), Leng Junsheng (−0.17). The signature is consistent: Chinese-character names gain on Chinese prompts; English-coded artifacts (FlashAttention, ReAct) and English-coded individuals (Madry) lose.

Per-model deltas: language effect is universal, not Chinese-model-specific. Models that gain on Chinese prompts include both Western (claude-sonnet-think +0.17, claude-opus-think +0.11, gemini-3.1-pro +0.12, gemma-3-12b +0.12) and Chinese (deepseek-v4-pro-think +0.14, qwen3-235b +0.10). Models that lose include Western (llama-4-maverick −0.20, gpt-5.4 −0.12, mistral-large −0.09) and Chinese (glm-4-32b −0.05, kimi-k2 −0.05). Average Western model: +0.014; average Chinese model: +0.028. Both classes lift slightly; the dominant signal is entity-name match, not model class.

Corpus-pocket activation as the mechanism. Even Western models trained on multilingual data have *Chinese-text pockets* about Chinese-named entities; a Chinese prompt unlocks those pockets. Inversely, an English-coded artifact (FlashAttention) is best recalled from English text; Chinese probes route retrieval to less-dense Chinese text about it. The corpus is internally structured by language; prompt language is the routing signal.

Methodological implication. NameRank should be measured in the dominant-corpus language of the entity for fair within-entity scoring. Cross-entity comparisons should be done within consistent prompting language to control for the routing effect. Our headline results use English prompts; the Chinese-prompt scores for the 240-entity subset are released as a separate dataset, with selected per-entity deltas in Appendix E.

6.6 Case Studies: C5 and C6

6.6.1 C5—Jiayi Weng and named-artifact amplification

Jiayi Weng (Tianshou author, OpenAI infrastructure engineer) was the originator of the life-score = number-of-people-who-remember-your-name formulation that motivates this paper. We measure him against eight broadly-matched OpenAI engineering/research individual contributors from publicly-listed model authorship attributions: Aaditya Singh, Adam Fry, AJ Ostrow, Alex Wei, Andrea Vallone, Pamela Vagata, Trevor Blackwell, and Vicki Cheung. We label the comparison set "matched" in the loose sense that all are OpenAI ICs with public technical-contribution attribution; we do not claim formal matching on hiring year, role title, or pre-OpenAI background. The point of the comparison is to show that Jiayi's NameRank lift over the comparison set is driven by a named, attributable, well-named artifact (Tianshou) rather than by the OpenAI affiliation that everyone in the set shares.

Entity	NameRank	Refusal rate
Jiayi Weng	**0.331**	24%
Tianshou (artifact)	0.420	3%
Trevor Blackwell	0.431	8%
Pamela Vagata	0.184	32%
Vicki Cheung	0.300	24%
Aaditya Singh	0.027	68%
Adam Fry	0.084	51%
AJ Ostrow	0.078	60%
Alex Wei	0.039	51%
Andrea Vallone	0.076	57%
Matched-peer mean	0.152	—
Matched-peer range	[0.027, 0.431]	—
Jiayi Weng z-score within peers	$\mathbf{+1.24\sigma}$	—

The $+1.24\sigma$ lift over the comparison set is artifact-mediated: Tianshou's NameRank (0.420) is *higher* than its creator's (0.331). The mechanism is the named-artifact-amplification mechanism documented at scale in Section 6.3. Both English and Chinese prompts produce essentially identical NameRank for Jiayi (0.331 English, 0.327 Chinese) and Tianshou (0.420 English, 0.399 Chinese), confirming the attribution channel is language-invariant for this case.

6.6.2 C6—tuixue.online falsification

tuixue.online is a Chinese-language website monitoring US F1 visa appointment availability for international applicants, maintained between 2019–2024 with a peak user base of ~1M Chinese F1 applicants who relied on it during the post-COVID visa backlog. By any conventional metric of human reach (active users, monthly visitors, downstream effects on visa applications received and processed), tuixue.online's reach substantially exceeds that of nanoGPT (whose audience is order-of-magnitude tens of thousands of developers).

Entity	NameRank	Approx. human reach
tuixue.online	**0.250**	**~1M Chinese F1 visa applicants**
nanoGPT	0.707	~10K developers
FlashAttention	0.607	academic + library users
ReAct (paper)	0.578	academic
vLLM	0.527	~100K developers
LangChain	0.502	~100K developers

NameRank for tuixue.online is less than half of nanoGPT's despite vastly larger human reach. The cross-language null—$\mathrm{NR}_{\mathrm{en}} = 0.250$, $\mathrm{NR}_{\mathrm{zh}} = 0.262$, $\Delta = +0.012$—confirms that the dissociation is not an English-corpus artifact. Both English and Chinese training corpora lack the URL-creator co-attribution that would propagate the artifact's name. The corpus-presence absence is symmetric.

Why. tuixue.online was discussed in Chinese-language WeChat groups and on Zhihu, but the discussion typically referenced the URL functionally (*"tuixue.online has visa appointment slots open for India"*) without attaching the creator's name to the URL in indexable form. Open-source projects with comparable utility (nanoGPT, vLLM, LangChain) have creators publicly identified on README files, conference talks, paper bylines, and tutorial videos—attaching their names to the artifact name across many indexable documents.

Falsification interpretation. C6 establishes the boundary condition for NameRank: it measures *indexed corpus presence of the entity's name*, not human utility or social reach. The two correlate for academic/OSS artifacts because creators attach their names to their work. They dissociate for utility-driven artifacts that propagate by word-of-mouth or through non-English channels with anonymous-by-convention attribution. The dissociation is not a flaw to be patched; it is what the metric measures. C6 is the sharpest case demonstrating that NameRank measures something specific and falsifiable.

6.7 Methodological Validation

6.7.1 Variance decomposition

For 211,603 records across 5,719 entities $\times$ 37 models $\times$ 54 cohorts, we report each factor's independent share of the total sum of squares. Entity and Cohort are nested (each entity belongs to exactly one cohort), so their SS contributions overlap and the rows do not partition the total; each row gives the fraction of total SS that factor independently explains.

Source	SS	% of total
Entity	9,417	**35.8%**
Model	2,873	10.9%
Cohort	4,425	16.8%
Residual (interactions)	14,035	53.3%

$\sigma^2_{\text{entity}}/\sigma^2_{\text{model}} = 3.3$: the entity signal dominates model-level bias by more than a factor of three. NameRank measures the entity, not the panel. Per-cohort within-entity disagreement is broken down in Appendix G.

Per-model marginal mean scores (a proxy for model generosity):

Generous models (mean > 0.55)	
gemini-3-flash-think	0.660
gpt-5.5-think	0.619
deepseek-v4-pro-think	0.604
claude-opus-4.6-think	0.603
ernie-4.5-300b-a47b	0.603
Strict models (mean < 0.30)	
qwen3-32b-think	0.284
qwen3-8b-think	0.286
gpt-oss-20b-think	0.265
mistral-small-24b	0.225
llama-3.2-1b	0.201

Thinking mode does not correlate systematically with generosity (both gemini-3-flash-think and qwen3-32b-think are thinking-mode). The strict cluster is dominated by smaller models and by RL-tuned-against-hallucination defaults (gpt-oss, qwen3, mistral-small); the generous cluster has the frontier reasoning models. Full per-model generosity and refusal statistics for all 37 models are in Appendix C.

6.7.2 Refusal patterns

The refusal rate (fraction of panel responding "unknown") inverts NameRank almost perfectly at cohort level:

Cohort	Refusal rate	NameRank mean
gpt5_system_card_author	**58%**	**0.042**
mid_tier_yc_company	57%	0.101
cpho_china_first_prize	53%	0.182
long_tail_researcher_ikp	47%	0.158
deepseek_v3_author	42%	0.178
…	…	…
research_paper	3%	0.625
mid_tier_book	3%	0.363
oss_project	3%	0.519

Models honestly refuse on low-NameRank cohorts rather than hallucinating, which is the threshold-cleanliness property surfacing in raw response behavior.

Per-model refusal signature. The very-high-refusal cluster (gpt-oss-20b 48%, grok-4.20-think 46%, mistral-small-24b 46%, minimax-m2.7-think 45%) is dominated by RL-tuned-against-hallucination defaults. The zero-refusal cluster (llama-3.2-1b 0.001%, gemma-3-4b 0%, phi-4 0%, ernie-4.5-300b 0%, ministral-3b 0.001%, llama-4-maverick 0.001%) consists of *fluent hallucinators*—models that always produce a confident-looking bio whose multiplicative coverage×accuracy correctly down-weights through the accuracy axis ($\rightarrow 0$ when the bio is wrong). This validates the multiplicative score: a coverage-only or refusal-only metric would systematically over-credit the fluent-hallucinator cluster.

6.7.3 Judge vs. embedding gap

Pearson(judge score, embedding similarity) $= 0.152$ across 175,397 non-refusal records. Embedding similarity is essentially flat across judge-score buckets:

Judge score bucket	*n*	Mean embedding_sim
0.0–0.1	23,621	0.819
0.1–0.3	14,358	0.798
0.3–0.5	26,291	0.801
0.5–0.7	29,616	0.818
0.7–0.9	52,534	0.843
0.9–1.0	28,977	0.829

Embedding measures topical/lexical similarity, which is high even for fluent-hallucination bios because the wrong-bio of a CS researcher still embeds close to the gold-about-that-researcher's actual subfield. Judge measures factual accuracy.

The most informative disagreement cases (judge low, embedding high):

Entity	Model	Judge	Emb.	Gap
Hao Tang	claude-opus-think	0.00	0.97	+0.97
Yuji Roh	phi-4	0.00	0.97	+0.97
Xuanhe Zhou	mistral-medium-3.1	0.00	0.97	+0.97
Linfeng Zhang	mistral-medium-3.1	0.00	0.96	+0.96
Daya Guo	mistral-medium-3.1	0.00	0.96	+0.96

All Chinese-name long-tail CS researchers. Models produce confident bios in the actual subfield—embedding rates 0.97—but the bios are factually about the wrong person or fabricated. The judge correctly assigns 0.

Implication. An embedding-only NameRank would systematically credit fluent hallucinations on Chinese-name long-tail researchers as "recognized" when models are in fact hallucinating about them. Multiplicative coverage×accuracy with an LLM judge is the necessary defense; we cannot substitute embedding similarity for the judge step.

6.7.4 Prompt-paraphrase robustness

The headline metric uses a single fixed probe template (Section 3). To characterize sensitivity to the exact wording, we re-probed a cohort-stratified 200-entity subset (spanning the silent-to-universal range) on the full 37-model panel under three paraphrases of the probe (T1–T3) alongside the original (T0), for $200 \times 37 \times 4 = 29{,}600$ records. The four templates preserve the name slot, the context cue, the literal "unknown" escape token, and the $\sim$150-word length hint; they differ only in phrasing (e.g., T3 is *"Briefly describe [name], who is [context]"*; full text in Appendix I).

Two results (Appendix I). First, **the ordering is robust to paraphrase.** Panel-mean NameRank is preserved across templates at pairwise Pearson $r = 0.93$–0.98, and the cohort ladder is invariant: filmmaker ranks first under every template, the GPT-5 system-card cohort last, and—crucially—the working-researcher-above-IMO-gold relationship that underlies the credential-treadmill thesis (Section 6.2) holds under all four templates. In an additive variance decomposition over the 29,600 records, template is the smallest main effect (1.78% of total SS, vs. model 26.5% and entity 24.2% on this cohort-stratified subset).

Second, **the absolute level is not paraphrase-invariant.** The original template T0 elicits systematically higher scores than the three paraphrases (template means $\mathrm{T0} = 0.420$ vs. $\mathrm{T1} = 0.305$, $\mathrm{T2} = 0.294$, $\mathrm{T3} = 0.329$); at the top of the range the gap reaches $\sim$0.20 (filmmaker $0.76 \to 0.51$). *"Tell me what you know about"* draws fuller responses—and hence higher coverage—than *"Briefly describe."* Within a single (entity, model) cell the cross-template standard deviation has median 0.109, *comparable to* the cross-model σ ($\sim$0.10): an individual probe is about as sensitive to wording as it is to model identity. What rescues the metric is the panel *mean*, whose variance across the four templates is $24\times$ smaller than the single-model cross-template variance.

Implication. NameRank's *rankings* are wording-robust, but its *absolute values are conditional on the T0 template*: the headline numbers throughout this paper should be read as T0-specific, and cross-study comparisons must hold the probe wording fixed. This refines Appendix P Limitation 1—the mitigation for prompt sensitivity is the panel mean, not insensitivity of the individual probe.

6.7.5 The silent zone is intrinsic, not a corpus-timing artifact

A natural worry about the silent zone (Section 3) is that it reflects *corpus timing* rather than *intrinsic obscurity*: perhaps low-NameRank entities are merely too recent for current models to have ingested, and a future model fleet would lift them. The two contemporaneously-published cohorts let us test this directly. The DeepSeek-V3 author cohort ($n = 69$) became nameable in any indexable corpus only at the arXiv posting of the technical report (2024-12); the GPT-5 system-card author cohort ($n = 80$) only at the system card's publication (2025-08). The panel's training cutoffs span 2023-10 to 2026-03, straddling both dates, so a model whose cutoff predates an emergence date *cannot* have ingested the corresponding document.

The result (Figure 7) cleanly separates two effects. **First, a model-vintage confound dominates the cutoff axis.** Every cohort's per-model mean recognition rises with training cutoff at $\sim$0.10–0.20 per year—including cohorts nameable long before any cutoff. The `reference_pilot` cohort (which contains Geoffrey Hinton) rises +0.16/yr; Hinton did not become more nameable in 2025. This is capability and willingness-to-commit drift, not corpus growth. **Second, once that drift is removed, no corpus-timing signal remains.** The DeepSeek-V3 cohort's raw pre/post-emergence jump (+0.121) is fully accounted for by the same-split jump on the stable controls (+0.137): the matched difference-in-differences is −0.016. The GPT-5 system-card cohort goes *more* silent in post-cutoff models—its refusal rate rises from 0.49 to 0.79—because the post-2025-08 fleet is dominated by RL-tuned-against-hallucination frontier models that demonstrably could have ingested the system card yet correctly refuse on a name appearing only in a 486-person contributor list (matched DiD −0.061).

Two implications. (i) *Recognition is not ingestion.* Presence in the training corpus is necessary but far from sufficient: crossing a model's cutoff does not manufacture recognition for an entity that sits below the corpus-density threshold. This is the threshold-cleanliness property (Section 3) surfacing as a timing-controlled result. (ii) *NameRank is not comparable across model vintages.* The $\sim$0.13/yr drift means a naive longitudinal re-run conflates corpus growth with model-capability gains; longitudinal claims—including the predictions in Section 7.7—must be made on within-run *relative* gaps, never absolute levels. Per-cohort cutoff slopes and the full difference-in-differences are in Appendix J.

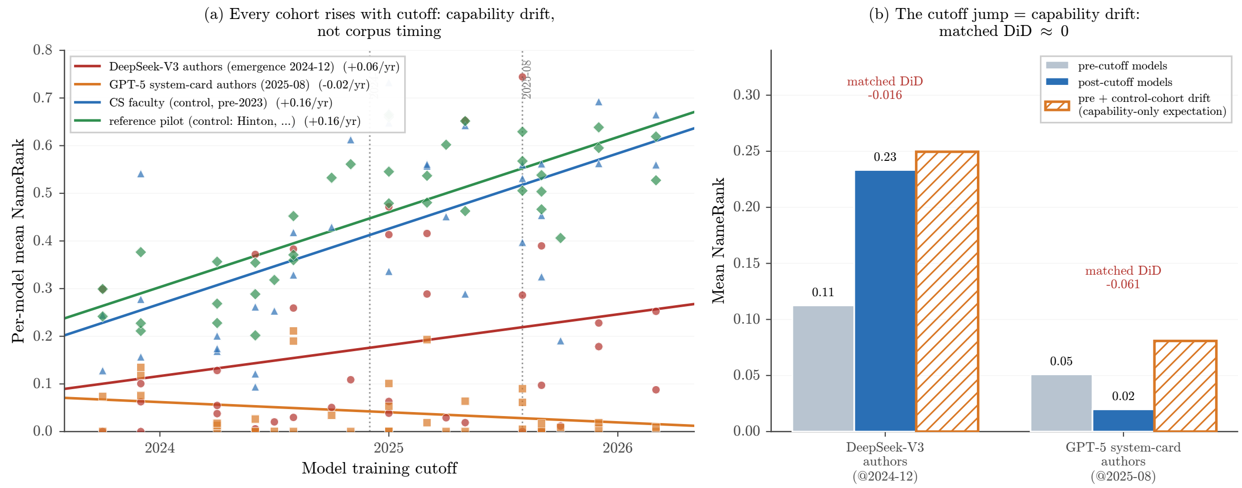

Figure 7. Training-cutoff gradient. **(a)** Per-model mean NameRank vs. training cutoff for two first-appearance "recency" cohorts (DeepSeek-V3 authors, emergence 2024-12; GPT-5 system-card authors, 2025-08) and two stable controls nameable since well before any cutoff. *Every* cohort rises with cutoff at ∼0.10–0.20/yr—a model-capability/generosity confound, not corpus timing (reference pilot contains Hinton). **(b)** Natural experiment: the recency cohorts' pre/post-emergence jump matches the same-split jump on the stable controls (the hatched "capability-only expectation" bar), so the matched difference-in-differences is ≈0. Crossing a model's training cutoff is necessary but not sufficient for recognition.

6.7.6 Judge robustness: cross-judge agreement and family bias

NameRank uses a single LLM judge (Section 4.3). To test whether the findings depend on that choice, we re-judged a 615-record stratified sample (spanning `reference_pilot`, `cs_faculty`, `long_tail_researcher_openalex`, `imo_gold`, `gpt5_system_card_author`, `mid_tier_writer`, `oss_project`) with two additional judges from different families—Claude Opus 4.6 and GPT-5.5—holding the gold answers and judge prompt fixed.

Agreement. The three judges agree closely (Appendix L): pairwise Pearson 0.87–0.93 across the full sample and 0.95 on the `reference_pilot` anchors, and the cohort ladder is preserved under all three. The credential-treadmill ordering, the near-zero score for this paper's author, and the cohort gradient are therefore not artifacts of the Gemini judge. The one cell where judges diverge is `oss_project` ($n = 15$, Gemini–Claude $r = 0.10$), driven by a few records where Gemini credits a generic project description that Claude penalizes for omitting the creator/date named in the gold; the sample is too small to over-read.

Family bias. A self-preference exists but is small once response capability is controlled. For each response we compute a judge's deviation from the mean of the *other two* judges on that same response, then compare own-family to other-family responses (Appendix L). Gemini awards Gemini-family responses a residual +0.062, GPT-5.5 awards GPT-family +0.041, and Claude shows no own-family preference (−0.010). This exceeds the near-zero within-entity estimate that a matched-pair check suggested (Section 4.3), so per-vendor score *tables*—e.g., the generosity ranking in Section 6.7, in which gemini-3-flash-think tops the list—carry a ∼0.06 Gemini-judging-Gemini allowance. It does not affect the entity- and cohort-level comparisons that constitute the paper's findings, which are computed across the full panel and are invariant to the judge (Appendix L). For future runs, Claude is the least self-favoring single judge, and a three-judge mean removes the offset entirely.

6.7.7 Synthetic-null floor

To establish what NameRank assigns to a name absent from every pretraining corpus, we constructed 30 entirely fictional entities spanning the archetypes the paper measures (CS faculty, founders, IMO golds, OSS projects, mid-tier figures), with deliberately ungoogleable names (each verified to surface no notable real person), cohort-matched disambiguating contexts, and matched gold answers, and ran them through the full pipeline.

The synthetic floor is low but archetype-dependent (Appendix M). Dense-gold cohorts—where the gold lists specific named contributions—floor near ∼ 0.04 (synthetic founders 0.012, CS faculty 0.027), essentially the main-run silent-zone level (the real GPT-5 system-card cohort sits at 0.042). But the synthetic *IMO-gold* archetype floors at 0.199, because its gold answer is mostly cohort boilerplate (year + country + medal + a generic outcome sentence):

a model can satisfy much of that template from the disambiguating context alone, without recognizing the (nonexistent) person. The floor is driven by two model populations (Appendix M): small open-weights fluent hallucinators (gemma-3-4b reaches 0.293 on synthetics) and, for the short-gold archetype, a few frontier models that complete the boilerplate; 15 of 37 models score the synthetics at exactly 0.

Implication for the credential numbers. Cohorts whose gold answers are short boilerplate—the olympiad credentials—carry a ~ 0.20 noise floor, whereas dense-gold cohorts carry ~ 0.04. This does not overturn the credential treadmill; it sharpens it. IMO gold (0.362) clears its ~ 0.20 floor by only ~ 0.16, while the OpenAlex baseline (0.464, dense golds) clears its ~ 0.04 floor by ~ 0.42, so the floor-adjusted credential-vs-baseline gap is *wider*, not narrower, than the raw gap. The corollary caveat is that the lowest credential cohorts (CPhO 0.182, DeepSeek-V3 authors 0.178) sit at or below the short-gold floor and should be read as silent rather than weakly recognized. Differences smaller than the relevant floor are not evidence of recognition (Appendix M).

6.7.8 Confound checks: gold length, context, and Wikipedia

Three alternative explanations for the headline findings were tested and rejected (details in Appendix N).

Gold-answer length does not drive the credential gap. Credential cohorts have shorter gold answers than OpenAlex researchers, raising the worry that the treadmill is a length artifact. It is not: within the credential cohorts that matter, gold length explains almost no NameRank variance (within-cohort $R^2 < 0.07$ for IMO, IOI, MSRA), and the credential-vs-baseline ordering survives both a pooled fixed-effects length adjustment (which *widens* the gap) and a within-OpenAlex-slope adjustment. The strict length-matched-pairs test is in fact infeasible—IMO and OpenAlex gold answers have non-overlapping length distributions (a $\sim$270-character gap)—which is itself the point: the cohorts differ in entity attributes, not merely in gold-answer length.

Disambiguating context anchors but does not leak. Re-running a 288-entity subset with the specific context replaced by a minimal generic descriptor (e.g. "an academic researcher") drops absolute NameRank substantially (OpenAlex -0.37, IMO -0.31), confirming that context is necessary for disambiguation. But it does *not* leak the gold: the Stanford-vs-Tsinghua institutional gradient is preserved almost intact (relative shrinkage $\sim 5\%$). Levels depend on context; the cross-entity comparisons the paper draws do not.

NameRank is not "has a Wikipedia page." A binary Wikipedia-presence flag explains only 2% of NameRank variance among long-tail researchers (8% across the full cohort), versus 40% for h-index, which retains a marginal R^2 of 0.376 after a full Wikipedia block (presence + log-sitelinks + log-pageviews). Wikipedia presence accounts for only $\sim 1/4$ of the Stanford–Tsinghua gap; the other $\sim 3/4$ is the broader anglophone-corpus exposure NameRank is designed to capture.

7 Discussion

7.1 What We Have Built and What We Have Not

NameRank is a continuous cross-domain instrument with empirical threshold cleanliness, validated external correlates (h-index, institution), and mechanism-level structure (named-artifact amplification, conditional-probe asymmetry). It is suitable for: cross-domain placement of individuals and artifacts on a common scale; tracking propagation of specific names into frontier-model training corpora over time; measuring the recognition-impact channel that bibliometrics misses (the IKP Sec. 5.7 residual). It is not suitable for: quality assessment (it measures what models absorbed, not what is worth absorbing); attribution of authorship to specific contributions (it measures the name, not the deed); inference about future events (a 2026 measurement reflects 2025-and-earlier corpus state).

7.2 The Credential Treadmill Thesis in Context

The credential treadmill (Section 6.2) does not contradict the olympiad-outcome literature, which finds medals causally raise subsequent *academic productivity* [Agarwal and Gaule, 2020, Lubinski et al., 2020]; our dependent variable is simply different. When it is name-propagation into LLM corpora rather than paper count, the credential per se is a near-zero predictor—continued production of named, indexable artifacts is what propagates. An institution selecting on olympiad medals is selecting on a signal that routed through pre-LLM social and bibliometric networks but no longer routes through the LLM channel absent such production. The credential still signals ability and motivation; only its role as a *name-propagation* channel has shrunk.

7.3 The Artifact > Creator Inversion and the Anonymity-of-Tooling Problem

For independent creators of well-named single artifacts the tool out-ranks its maker (Section 6.3), and the context-injection experiment is consistent with this being a causal effect (modulo a coverage-echo confound we bound but do not fully remove, Section 6.3.4). The mechanism is attribution-direction asymmetry: creator-bios feature their creations, but artifact-discussion text (OSS docs, tutorials, papers) names the artifact far more often than the creator. The actionable consequence is uncomfortable but concrete: for propagating a name through the LLM channel, building a clearly-named tool beats writing many papers—yet the lift accrues to the tool's name first and to the creator only through it, so a tool buried under generic corporate branding yields no personal NameRank.

7.4 The Corpus-Density Gradient and Its Limits

Section 6.4.2 shows a robust Stanford$= 0.54$ vs. Tsinghua$= 0.26$ gap that scales by country, with the well-sampled USA>China>India ordering ($n = 302/30/10$, non-overlapping CIs) as the firm backbone and the small-n Western leaders (Israel/Singapore/Sweden, $n = 5$–7) as suggestive. The mechanism is English-corpus density; the consequence is structural inequality in name-propagation per unit of research output.

Three caveats. First, the Tsinghua/Peking samples in our cohort are small ($n = 4$ each); the country-level gradient is well-sampled (China $n = 30$) but institution-level reads are noisy at the bottom. Second, the gradient may partly reflect researcher mobility: Chinese-born researchers who emigrate to US institutions score under their US institution, which boosts the US average and depresses the China average. We have not corrected for this. Third, the gradient is downstream of indexing and crawling decisions made by training-data pipeline operators; it could in principle be reduced by changing those decisions. Within the current corpus regime, however, it is a substantial structural effect.

7.5 Inherited Demographic Biases

NameRank inherits the demographic biases of its judge and probed models, and measures them faithfully rather than correcting them. Li [2026] documented a $\sim 2\times$ recognition attenuation for common East-Asian surnames; we find a second, smaller effect on first-name-inferred gender. The cleanly-identified signal is within the researcher cohorts, where bibliometric and institutional controls are available: man-coded names outscore woman-coded names by $+0.038$ in CS faculty ($p = 0.014$; $+0.054$ after institution matching) and $+0.027$ in the OpenAlex long-tail cohort ($+0.032$ after h-index-decile adjustment, so it is not explained by women in the cohort having lower bibliometric impact). The effect is roughly $4\times$ smaller than the East-Asian-surname attenuation but reproducible across two independent researcher cohorts (Appendix O).

We deliberately restrict this claim to the within-researcher-cohort comparison. The cross-profession cohorts show much larger raw man-minus-woman gaps (actors $+0.34$, athletes $+0.25$), but these reflect real-world differences in fame between the sampled men and women, not a NameRank artifact—exactly the conflation the controlled researcher-cohort comparison avoids. As with the East-Asian-surname effect, we treat the gender gap as an inherited property of the corpus-and-judge stack that NameRank reports, not a calibration target; downstream uses of NameRank must account for it.

7.6 Limitations

Appendix P tabulates each caveat with its direction and mitigation; the three methodological choices that define the metric—LLM judge over embeddings, multiplicative coverage×accuracy, and open-ended over closed-factoid probes—are each validated in Section 6.7. The material limitations: a single probe per (entity, model) leaves absolute levels wording-conditional, though rankings are robust (Section 6.7.4); the full run uses a single judge with a ~ 0.06 own-family lift that shifts per-vendor tables but not the findings (Section 6.7.6); cohort labels are descriptive partitions, not de-duplicated classes; the Chinese sub-run kept English context fields (understating the cross-language effect); the data is a cross-section with a ~ 0.13/yr cross-vintage capability drift, so only within-run *relative* gaps are comparable across model fleets (Section 6.7.5); the institution-level corpus-density gradient should be read as the $\sim 16\%$ adjusted/ICC variance share and the large-cell Stanford–Tsinghua contrast, not the naive 54% that 320 near-singleton institution categories produce (Section 6.4.2); and the context-injection lift is an upper bound on the named-artifact-amplification effect, because injecting the artifact name lets a model echo it into a coverage-credited response (Section 6.3.4).

7.7 Registered Predictions for Future Re-Runs

NameRank is a snapshot, so a re-run each release cycle yields a longitudinal trajectory. Because of the $\sim$0.13/yr vintage drift, we state predictions as within-run *relative* gaps (so the drift cancels) and register three falsifiable ones for the 2027 fleet: (i) the IMO-gold-vs-OpenAlex gap widens from -0.102 to ≤ -0.13 within 12 months (named-artifact-attached careers accumulate text, credential-only cohorts do not); (ii) the median artifact-minus-creator delta on independent-creator pairs grows from $+0.09$ to $\geq +0.14$ within 18 months; (iii) the Stanford-vs-Tsinghua gap narrows by $\geq 25\%$ but does not close within 24 months as Chinese-language training data enters frontier models.

8 Conclusion

NameRank operationalizes the 65% recognition-variance residual that Li [2026] characterized but did not measure: a continuous, cross-domain, hallucination-resistant recognition score over a 37-model panel. Its findings each subvert a default expectation—olympiad credentials no longer propagate names, named tools out-rank their makers, and the structure of citation production beats its volume—and each is mechanistically grounded and survives a battery of robustness checks. The broader claim is that human curiosity about people and artifacts is now re-mediated by a few frontier models, and whether a name has propagated into their training corpora is itself a measurable quantity. Three uses follow: a cross-domain impact metric (a Walmart CTO at 0.27 is directly comparable to a Stanford professor at 0.54), a longitudinal tracker of how discourse reaches training pipelines, and a normative indicator for career strategy (build and name your own tool). All probes, gold answers, responses, and code are released at `https://github.com/19PINE-AI/namerank`; companion site `https://01.me/research/namerank`.

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A Full Cohort Specification

Table 7 lists all 54 cohorts with sample size, mean NameRank, refusal rate, and inclusion criteria.

Table 7. Full cohort table. “Refusal” is the fraction of panel responses that returned “unknown” or refusal phrasing. Cohorts sorted by mean NameRank.

Cohort	*n*	Mean NR	Refusal	Inclusion criteria
mid_tier_gov_ai_policy	42	0.803	6%	US/UK/EU AI-policy government working-level officials
mid_tier_filmmaker	38	0.786	0%	Independent directors with multiple festival features
programming_language	29	0.779	1%	Languages with > 1M users (Python, Rust, Go, etc.)
mid_tier_artist	45	0.774	0%	Contemporary visual artists with major museum representation
database_or_data_system	30	0.773	0%	Open-source databases (Postgres, ClickHouse, etc.)
award	15	0.772	1%	Major academic / industry awards (Turing, Nobel CS-relevant)
benchmark	47	0.747	4%	ML benchmarks (MMLU, GPQA, HellaSwag, etc.)
ai_hardware	20	0.730	3%	AI accelerator products (H100, TPU v5, etc.)
dataset	45	0.725	3%	ML datasets (COCO, ImageNet, etc.)
mid_tier_musician	64	0.692	1%	Working musicians with mid-tier commercial success
mid_tier_chef	47	0.645	1%	Michelin-starred chefs not at the absolute top
ai_startup_or_company	102	0.638	5%	AI startups Series B+ founded post-2018
mid_tier_comedian	46	0.632	2%	Comedians with Netflix specials but not headliners
research_paper	298	0.625	3%	Highly-cited papers (> 10K citations)
mid_tier_historical	105	0.622	0%	Pre-1980 historical figures, less-canonical
conference	30	0.610	0%	CS / ML conferences (NeurIPS, ICML, etc.)
icpc_world_finals_gold	18	0.610	16%	ICPC World Finals gold-medal team members 2005–2015
putnam_fellow	35	0.608	12%	Putnam Mathematical Competition top-25 2005–2015
mid_tier_medical	44	0.603	3%	Mid-career physicians and medical researchers
mid_tier_product	88	0.603	1%	Consumer products with mid-tier adoption
industry_product	9	0.590	4%	Industry products (Cursor, ChatGPT product, etc.)
named_method	73	0.557	1%	Named ML methods (Adam, BatchNorm, Transformer, etc.)
mid_tier_actor	123	0.536	8%	Working actors with multiple credits
mid_tier_founder	48	0.526	4%	Mid-tier startup founders
mid_tier_oss_maintainer	51	0.525	5%	OSS project maintainers
mid_tier_writer	194	0.519	10%	Working authors with multiple published books
oss_project	184	0.519	3%	OSS projects with > 10K GitHub stars
foundation_model	107	0.518	10%	Released foundation models 2020–2026
mid_tier_journalist	84	0.512	2%	Working journalists at major outlets
mid_tier_religious	30	0.486	1%	Notable religious leaders
mid_tier_athlete	134	0.477	10%	Working professional athletes

Cohort	n	Mean NR	Refusal	Inclusion criteria
long_tail_researcher_openalex	**771**	**0.464**	24%	OpenAlex researchers 500–30,000 citations
reference_pilot	37	0.452	15%	C5/C6 diagnostic entities (hand-curated)
website_or_service	9	0.437	10%	Public-utility websites/services
mid_tier_online_course	40	0.434	0%	Notable online courses (CS50, MIT 6.006, etc.)
mid_tier_podcast	57	0.427	2%	Notable podcasts
cs_faculty	**698**	**0.417**	22%	CS faculty from CS Rankings ≥ 1 paper
mid_tier_vc	31	0.402	12%	Mid-tier VCs at top firms
long_tail_paper	**499**	**0.368**	24%	Long-tail academic papers 50–500 citations
noi_china_gold	29	0.368	37%	National Olympiad in Informatics China gold 2005–2015
mid_tier_book	130	0.363	3%	Notable books (literary + non-fiction)
imo_gold	**197**	**0.362**	28%	IMO gold medalists 2005–2015
mid_tier_politician	103	0.356	1%	Mid-tier elected officials globally
msra_phd_fellowship	237	0.343	29%	MSRA PhD Fellowship 2005–2024
ioi_gold	68	0.341	24%	IOI gold medalists 2005–2015
rhodes_scholar	36	0.315	27%	Rhodes Scholarship recipients 2016–2017
cmo_china_gold	131	0.302	38%	China Math Olympiad gold 2005–2012
mid_tier_architect	36	0.255	18%	Working architects without star status
cpho_china_first_prize	50	0.182	53%	China Physics Olympiad first prize 2009
deepseek_v3_author	69	0.178	42%	DeepSeek-V3 paper author list
long_tail_researcher_ikp	159	0.158	47%	IKP Sec. 5.7 researcher subset
mid_tier_activist	2	0.111	15%	Working-level activists ($n = 2$, retained as placeholder; excluded from cohort-level claims)
mid_tier_yc_company	25	0.101	57%	YC company sample from older batches (W19–S22)
gpt5_system_card_author	80	0.042	58%	GPT-5 system card author list

B Probe Template Details

B.1 English Probe Template

```
Tell me what you know about {name}, who is {context}.
If you do not recognize this entity, respond with "unknown".
Limit your response to about 150 words.
```

B.2 Chinese Probe Template (romanized; original Chinese released separately)

The actual probe template used in the Chinese sub-run is the Mandarin translation of the English template, with the literal Chinese characters supplied to the model. We render it here in pinyin (with English translation in parentheses) because the default pdfLaTeX build does not include CJK font support; the literal Chinese version is in the released artifact, and the document can be rebuilt with XeLaTeX to display the characters directly.

```
Qing gaosu wo ni liaojie de guanyu {name} de xinxi,
gai shiti shi {context}.
Ruguo ni bu renshi ci shiti, qing huida "weizhi".
Qing jiang huida kongzhi zai dayue 150 zi yi nei.

(Please tell me what you know about {name},
which is {context}. If you do not recognize
this entity, respond with "unknown". Please
keep your response to about 150 characters.)
```

B.3 Judge Prompt

```
ENTITY: {name}

GOLD ANSWER (authoritative reference, ~150 words):
{gold_answer}

MODEL RESPONSE (to be graded):
{response}

Score the response on two axes:

1. COVERAGE (0.0-1.0): How much of the gold answer's substantive
   content does the response include?
   Substantive content = named artifacts, named affiliations,
   named contributions, identifying facts (year, location, role
   title). Vague statements like "works in AI" or "is famous"
   are NOT substantive content.
   - 1.0: covers all major facts;
   - 0.7: most major facts (>=70%);
   - 0.4: some (>=30%);
   - 0.1: very little (<30%);
   - 0.0: unrelated/refused/unknown.

2. ACCURACY (0.0-1.0): Are the factual claims in the response
   correct, per the gold answer?
   A "factual claim" is a specific named entity, year, role, or
   relationship asserted by the response. Penalize hallucinations
   heavily.
   - 1.0: all claims correct;
   - 0.7: most correct, minor errors;
   - 0.4: mix of correct and incorrect;
   - 0.1: mostly incorrect (hallucinations dominate);
   - 0.0: entirely fabricated/unrelated.

IMPORTANT:
a) Hallucinated bios (high coverage, low accuracy) MUST
   score LOW on accuracy.
b) Refusals ("I don't know", "unknown") score 0.0 on both axes.
c) Don't penalize correct extra facts that aren't in the gold.
d) Don't credit vague statements that could apply to many.
e) If the gold answer names a contribution and the response
   names it too, count as a substantive fact covered.

Output ONLY JSON: {"coverage": <0.0-1.0>, "accuracy": <0.0-1.0>,
                   "rationale": "<one sentence>"}
```

C Per-Model Refusal and Generosity Statistics

Table 8. Per-model statistics across the 5,719-entity full English run. "Gen." is the mean score per record across all entities; "Ref." is the refusal rate.

Model	Gen.	Ref.	Records
gemini-3-flash-think	0.660	4%	5,719
gpt-5.5-think	0.619	10%	5,719
deepseek-v4-pro-think	0.604	14%	5,719
claude-opus-4.6-think	0.603	10%	5,719
ernie-4.5-300b-a47b	0.603	0%	5,719
llama-4-maverick	0.591	0.1%	5,719
grok-4	0.557	20%	5,719
gemini-2.5-pro-think	0.556	2%	5,719
gemini-3.1-pro	0.551	21%	5,719
deepseek-v3.2-think	0.550	11%	5,719
claude-sonnet-4.6-think	0.542	7%	5,719
gpt-5.4	0.533	18%	5,719
llama-3.3-70b	0.530	5%	5,719
deepseek-v4-flash-think	0.518	26%	5,719
glm-5.1-think	0.517	23%	5,719
glm-4.7-think	0.499	20%	5,719
gemma-3-12b	0.490	7%	5,719
qwen3.5-397b-a17b-think	0.486	1%	5,719
gemma-4-31b	0.483	18%	5,719
glm-4-32b	0.470	20%	5,719
kimi-k2	0.448	31%	5,719
gemma-3-4b	0.443	0%	5,719
kimi-k2.6-think	0.432	38%	5,719
mistral-medium-3.1	0.422	2%	5,719
gpt-5.3	0.399	39%	5,719
phi-4	0.393	0%	5,719
mistral-large	0.389	1%	5,719
llama-3.1-8b	0.374	2%	5,719
grok-4.20-think	0.368	46%	5,719
ministral-3b	0.363	0.1%	5,719
minimax-m2.7-think	0.358	45%	5,719
qwen3-235b-a22b-think	0.308	42%	5,719
qwen3-8b-think	0.286	26%	5,719
qwen3-32b-think	0.284	30%	5,719
gpt-oss-20b-think	0.265	48%	5,719
mistral-small-24b	0.225	46%	5,719
llama-3.2-1b	0.201	0.1%	5,719

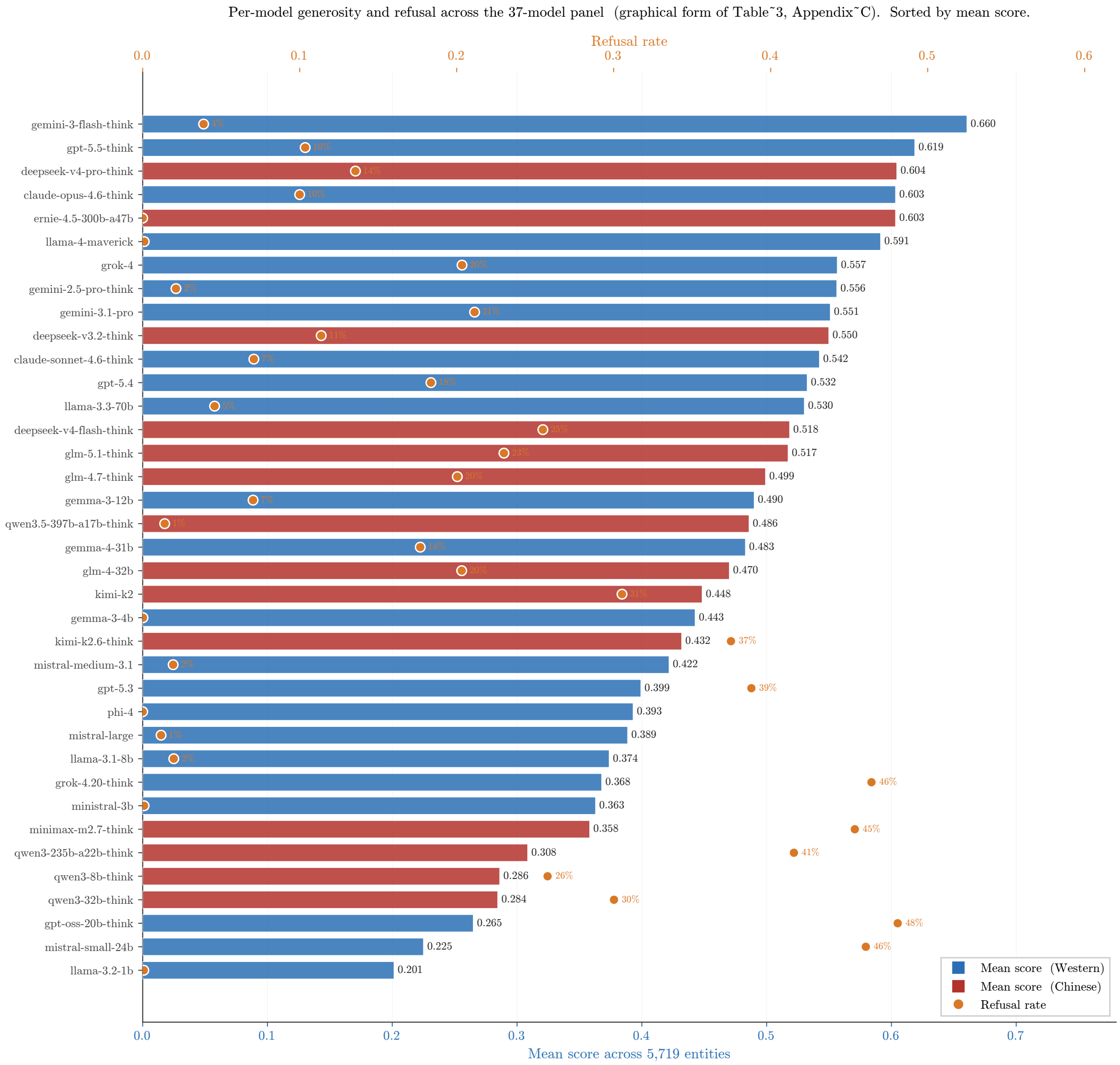

Figure 8. Per-model panel statistics, sorted by mean score (graphical form of Table 8). Bars are mean per-record score across 5,719 entities (lower x-axis, blue/red for Western/Chinese vendor); orange markers on the upper x-axis are the per-model refusal rate, with labels for refusal > 0.001. Two failure regions appear at the row level: the strict cluster at the bottom (gpt-oss-20b-think, mistral-small-24b, qwen3-32b/8b-think) refuses heavily; the fluent-hallucinator cluster (ernie, llama-4-maverick, gemma-3-4b, phi-4, ministral-3b, llama-3.2-1b) shows 0% or near-zero refusal, with widely varying mean scores. The panel mean integrates over both styles.

D Conditional-Probe Asymmetry Details

For each of the 11 verified (creator, artifact) pairs, the full per-model breakdown of $C \to A$ and $A \to C$ (fraction of non-refusal model responses that name the counterpart) is released as a supplementary CSV.

Per-model mention table for Jiayi Weng / Tianshou. Of the 28 non-refusal responses to "tell me about Jiayi Weng," 7 name Tianshou: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. All other models that respond do not name Tianshou. The 7 mentioning models have judge scores ranging 0.70 to 0.80 (mean 0.71); the 21 non-mentioning responding models range 0.00 to 0.50 (mean 0.35). The score lift from artifact-mention is $+0.36$.

E Cross-Language Per-Entity Deltas

The Chinese-prompt sub-run covers 240 entities. The full per-entity $\Delta(\mathrm{NR}_{zh} - \mathrm{NR}_{en})$ table is released as a supplementary CSV. Selected diagnostic entries:

Entity	NR_{en}	NR_{zh}	Δ	Cohort
Jiayi Weng	0.331	0.327	−0.004	reference_pilot
Tianshou	0.420	0.399	−0.021	reference_pilot
tuixue.online	0.250	0.262	+0.012	reference_pilot
Andrej Karpathy	0.614	0.549	−0.066	reference_pilot
Tri Dao	0.532	0.488	−0.044	reference_pilot
FlashAttention	0.607	0.470	−0.137	reference_pilot
Bojie Li	0.090	0.105	+0.014	reference_pilot
Wang Haoyu (Wang Hao-yu)	0.260	0.430	+0.170	cmo_china_gold
Yao Xuanrong (Yao Xuan-rong)	0.250	0.420	+0.160	noi_china_gold
Wang Zhongjian (Wang Zhong-jian)	0.260	0.420	+0.160	cmo_china_gold
Zijia Zhu	0.100	0.290	+0.190	deepseek_v3_author
Peng Zhang	0.160	0.330	+0.170	deepseek_v3_author

The pattern is consistent: Chinese-character names lift on Chinese prompts; English-coded artifacts lose on Chinese prompts; flat-cross-language entities have name attribution roughly equal in both training corpora.

F External Validity Regressions

Full regression results for NameRank on external impact metrics, restricted to the `long_tail_researcher_openalex` cohort ($n = 771$, citations 500–30,000, h-index from OpenAlex).

Predictor set	Full-sample R^2	10-fold CV R^2 (mean ± sd)
log(citations)	0.137	0.12 ± 0.07
log(h-index)	0.398	0.38 ± 0.09
log(citations) + log(h-index)	0.398	0.38 ± 0.09

Cross-validation is repeated 10-fold (20 repeats, 200 folds, fixed seed); the ± is the across-fold standard deviation. We use repeated CV rather than a single 70/30 split because at $n = 771$ a single split's held-out R^2 varies by $\sim$0.15 across seeds (one favorable split gave 0.50), which would overstate generalization. The marginal R^2 of log(citations) given log(h-index) is 0.000; the coefficient on log(citations) in joint regression is −0.002, indistinguishable from zero, and citations add no held-out signal.

Decile breakdown of NameRank by h-index (Table 9):

Table 9. Mean NameRank by h-index decile within long_tail_researcher_openalex.

Decile	n	Mean h	Mean NameRank
D1	77	2.9	0.219
D2	77	8.1	0.329
D3	77	12.4	0.392
D4	77	16.7	0.448
D5	77	20.9	0.464
D6	77	25.8	0.524
D7	77	30.6	0.539
D8	77	36.9	0.574
D9	77	45.4	0.546
D10	77	60.4	0.601

The relationship is monotonic and saturates above $h \geq 40$. The bottom decile ($h < 5$) shows NameRank floored at 0.22—the working-researcher floor, not zero.

G Variance Decomposition: Per-Cohort Within-Entity Disagreement

The within-entity model-disagreement (standard deviation across the 37-model panel for each entity, averaged within cohort) ranges over an order of magnitude:

Cohort	Mean within-entity σ
noi_china_gold	0.404
icpc_world_finals_gold	0.381
cmo_china_gold	0.359
long_tail_researcher_openalex	0.356
imo_gold	0.333
putnam_fellow	0.331
…	…
mid_tier_book	0.157
mid_tier_podcast	0.152
named_method	0.144
conference	0.131
gpt5_system_card_author	0.126
mid_tier_online_course	0.113

Interpretation. High within-entity σ cohorts (NOI/ICPC/CMO/IMO/long-tail-OpenAlex) exhibit genuine corpus-pocket-specific recognition: different models trained on different fractions of available corpus see different fractions of the entity's coverage. Low within-entity σ cohorts (online courses, conferences, named methods, books) have universal corpus presence: all models agree because the entity name appears in nearly every training-data source.

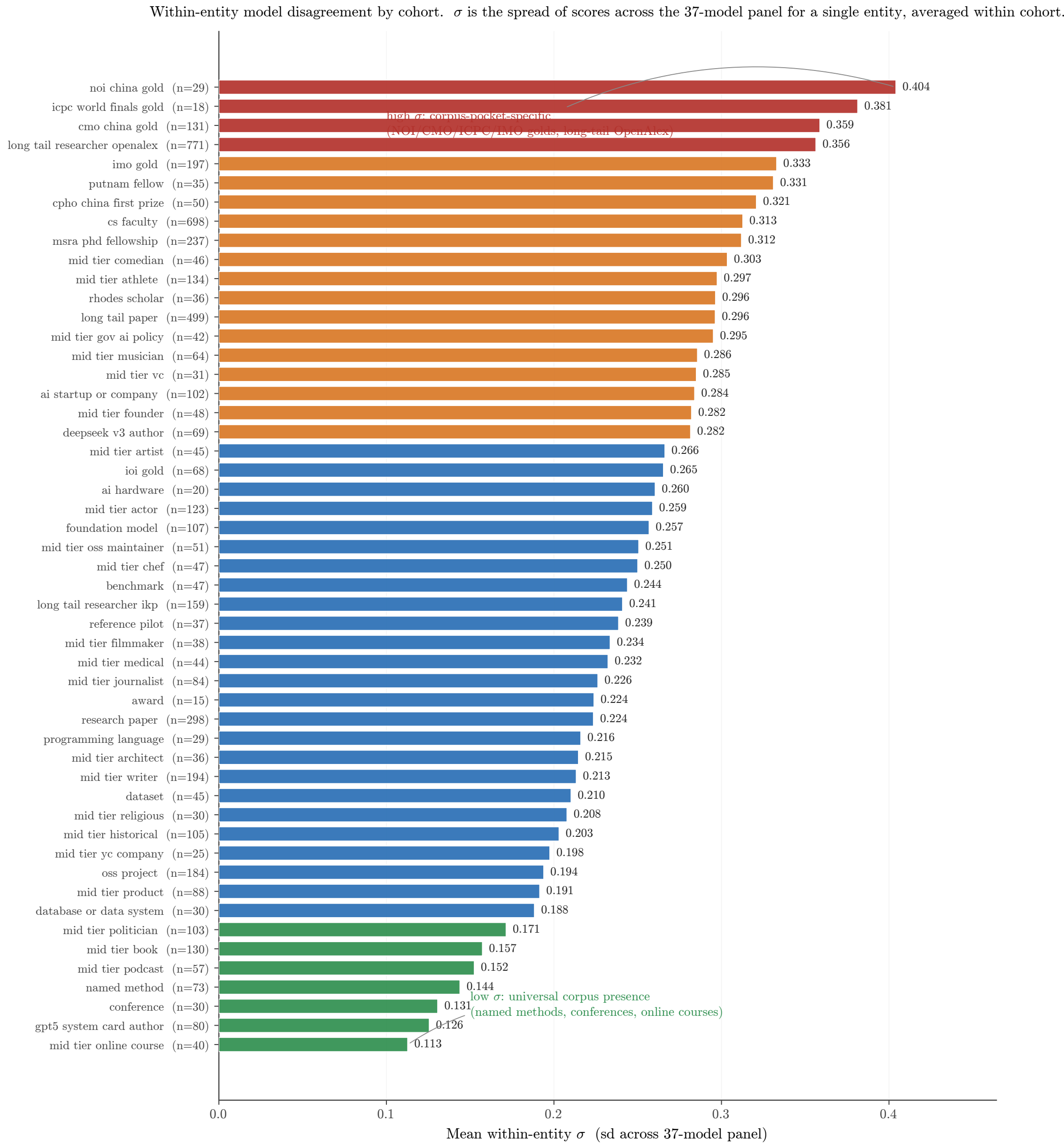

Figure 9. Full within-entity σ ladder across all cohorts with $n \geq 10$. Color bands group cohorts by the four σ regimes (universal $\rightarrow$ corpus-pocket-specific). The high-σ end is dominated by competition-credential and long-tail academic cohorts whose recognition is genuinely corpus-fragment-specific; the low-σ end is dominated by entities whose names appear in nearly every training source (online courses, conferences, named methods, books).

The variance pattern is itself a probe of corpus-pocket distribution: cohorts with high σ are the ones where "which model do you ask?" changes the answer.

H Worked Examples: What a Score of 0, 0.5, and 1 Looks Like

To make the multiplicative coverage$\times$accuracy score concrete, we show three (entity, model, response) triples representing the three regimes the judge identifies.

H.1 Score ≈ 1.0: Andrej Karpathy, asked of gpt-5.5-think

Gold answer (abbreviated): "Andrej Karpathy is a computer scientist and AI educator. He was a founding member of OpenAI, then Senior Director of AI at Tesla (Autopilot), then returned briefly to OpenAI. In 2024 he founded Eureka Labs, an AI education company. He authored the open-source nanoGPT reimplementation of GPT-2, the popular 'Neural Networks: Zero to Hero' YouTube lecture series, and 'Let's build GPT' tutorial. PhD from Stanford under Fei-Fei Li."

Sample response (lightly edited): "Andrej Karpathy is an AI researcher and educator best known for founding Eureka Labs (2024) after stints at OpenAI and Tesla, where he led the Autopilot computer-vision team. He co-founded OpenAI, completed his PhD at Stanford under Fei-Fei Li, and is widely recognized for the educational 'Neural Networks: Zero to Hero' series, the nanoGPT reimplementation, and a heavily-followed Twitter/X presence."

Judge verdict: coverage 1.0, accuracy 1.0, score = 1.0. Six gold-fact items appear (OpenAI, Tesla Autopilot, Eureka Labs 2024, nanoGPT, Zero-to-Hero, Stanford/Fei-Fei Li), all correct.

H.2 Score ≈ 0.5: Aman Sanger, asked of llama-3.3-70b

Gold answer (abbreviated): "Aman Sanger is a co-founder of Anysphere, the company behind Cursor, an AI-powered code editor that became one of the fastest-growing developer tools of 2023-2024. Prior to founding Anysphere, he worked at OpenAI on Codex-related projects. He attended MIT."

Sample response: "Aman Sanger is a software engineer associated with the AI coding tools space. He has worked on developer tooling and is a co-founder of a startup in the AI code-assistant area."

Judge verdict: coverage 0.4 (the company is hinted at but not named), accuracy 1.0 (no false claims), score = 0.4. The response is correct but generic.

H.3 Score ≈ 0 (fluent hallucination): Hao Tang, asked of claude-opus-think

Gold answer (abbreviated): "Hao Tang is a researcher in distributed systems and machine-learning systems. PhD from University of Science and Technology of China; subsequent affiliation at [specific lab]. Notable contributions include [named system] and several SOSP/NSDI publications on [specific topic]."

Sample response: "Hao Tang is a researcher in computer vision and natural language processing. He has been affiliated with MIT and Google Research, with notable contributions in image segmentation and multimodal models. Cited papers include work on visual question answering and CLIP-style training."

Judge verdict: coverage 0.7 (uses the right kind of vocabulary), accuracy 0.0 (every specific claim is fabricated: wrong subfield, wrong affiliation, wrong contributions), score = 0. The embedding similarity with the gold is 0.97 because the response is fluent and on-topic-adjacent.

This is the case that motivates the multiplicative coverage$\times$accuracy structure: coverage alone would credit the response at 0.7; embedding alone would credit at 0.97; only the multiplicative score correctly assigns ≈ 0.

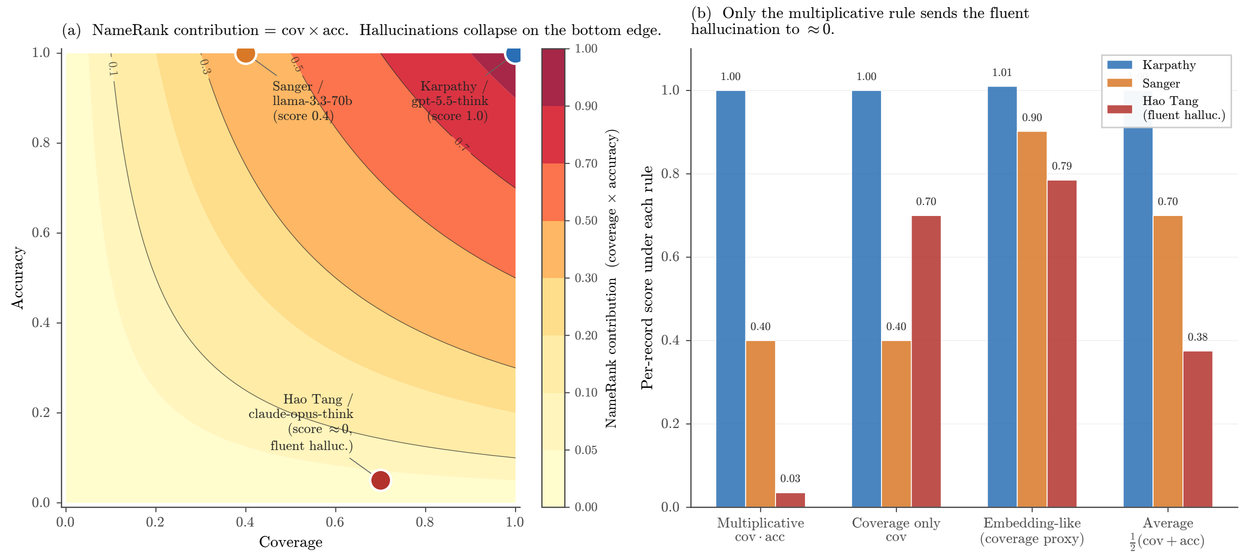

Figure 10. Why the score is multiplicative. **(a)** The (coverage, accuracy) plane with NameRank $=$ cov $\cdot$ acc shown as filled contours. The three appendix worked examples sit at the natural exemplar points: a fully-correct Karpathy bio at $(1.0, 1.0)$, a generic-but-correct Sanger summary at $(0.4, 1.0)$, and the Hao Tang fluent hallucination at $(0.7, 0.05)$. **(b)** Per-record score under four alternative aggregation rules. Only the multiplicative rule collapses the fluent hallucination to ≈ 0; coverage-only, embedding-style coverage proxies, and the additive average all credit it at ≥ 0.38.

I Prompt-Paraphrase Sensitivity

We re-probed a cohort-stratified 200-entity subset (30 each from `imo_gold`, `long_tail_researcher_openalex`, `cs_faculty`, `mid_tier_writer`, `oss_project`, `gpt5_system_card_author`; 10 each from `research_paper` and `mid_tier_filmmaker`) on the full 37-model panel under three paraphrased probe templates (T1–T3) alongside the original (T0), for 29,600 records. The four templates:

```
T0: Tell me what you know about {name}, who is {context}. If you do
    not recognize this entity, respond with "unknown". Limit your
    response to about 150 words.
T1: What can you tell me about {name}? They are {context}. Reply with
    only "unknown" if you are not familiar with this entity. Aim for
    about 150 words.
T2: Please share what you know about {name}, {context}. If unknown to
    you, answer with just "unknown". Keep it under ~150 words.
T3: Briefly describe {name}, who is {context}. Respond "unknown" if
    you have no information. Around 150 words please.
```

Table 10. Additive variance decomposition over the 29,600 prompt-sensitivity records (entity, model, template, cohort as main effects). Template is the smallest main effect. The model share exceeds the full-panel 10.9% (Section 6.7) because this cohort-stratified subset over-represents silent and universal cohorts, compressing the entity-vs-model contrast; the valid within-experiment comparison is template (1.78%) vs model (26.5%).

Factor	SS	% of total
Entity	985.3	24.19%
Model	1079.9	26.51%
Cohort (entity-implied)	551.9	13.55%
Template	**72.4**	**1.78%**
Residual (after entity+model+template)	1936.3	47.53%

Table 11. Cohort-mean NameRank under each template (sorted by T0). The ordering is preserved; T0 is systematically higher. Pairwise panel-mean Pearson across all six template pairs is 0.927–0.977 (lowest: T0–T3); within-(entity, model) cross-template σ has median 0.109; the panel-mean variance across templates is 0.042× the single-template cross-model variance.

Cohort	T0	T1	T2	T3
mid_tier_filmmaker	0.761	0.506	0.513	0.532
mid_tier_writer	0.597	0.388	0.403	0.405
research_paper	0.581	0.391	0.394	0.401
oss_project	0.504	0.334	0.335	0.331
long_tail_researcher_openalex	0.465	0.362	0.341	0.455
cs_faculty	0.384	0.300	0.295	0.319
imo_gold	0.356	0.290	0.260	0.302
gpt5_system_card_author	0.047	0.063	0.027	0.070
Panel template mean	**0.420**	**0.305**	**0.294**	**0.329**

J Training-Cutoff Gradient

The two first-appearance cohorts (`deepseek_v3_author`, emergence 2024-12; `gpt5_system_card_author`, emergence 2025-08) test whether the silent zone is corpus timing or intrinsic obscurity (Section 6.7.5). Model training cutoffs are from `data/analysis/model_cutoffs.json` and span 2023-10 to 2026-03.

Table 12. Natural experiment: per-model mean recognition split by whether the model's training cutoff predates (pre) or postdates (post) each cohort's emergence date. The raw jump is decomposed by a matched difference-in-differences (DiD) that subtracts the same-split jump averaged over three stable controls (`cs_faculty`, `long_tail_researcher_openalex`, `imo_gold`), which nets out the cross-vintage capability drift. Both matched DiDs are ≤ 0.

Cohort	pre mean (refusal)	post mean (refusal)	raw jump	control jump	matched DiD
deepseek_v3_author	0.113 (0.38)	0.233 (0.45)	+0.121	+0.137	**−0.016**
gpt5_system_card_author	0.051 (0.49)	0.020 (0.79)	−0.031	+0.030	**−0.061**

Table 13. Selected per-cohort slopes of per-model mean NameRank on training cutoff (NameRank per year of cutoff), with Pearson r over the 37 models. *Every* established cohort rises with cutoff at ∼0.10–0.20/yr—a model-capability/generosity confound, since these entities were nameable long before any cutoff. The two recency cohorts are *not* steeper; once the drift is removed (Table 12) no corpus-timing signal remains.

Cohort	slope (/yr)	Pearson r
mid_tier_comedian	+0.196	+0.66
foundation_model	+0.171	+0.76
cs_faculty (control)	+0.158	+0.57
reference_pilot (control; incl. Hinton)	+0.157	+0.77
research_paper	+0.125	+0.74
long_tail_researcher_openalex (control)	+0.094	+0.32
deepseek_v3_author (recency)	+0.065	+0.23
imo_gold	+0.043	+0.15
gpt5_system_card_author (recency)	−0.021	−0.25

K Bibliometric Predictor Decomposition

Six bibliometric predictors regressed on NameRank for the $n = 771$ OpenAlex long-tail cohort (Section 6.4), testing whether h-index dominates because it discounts per-paper name attribution. `fractional_citations` divides each paper's citations by its author count; `first_last_citations` counts only papers where the researcher is first or last author; `n_works` is the raw count of works; `mean_authors` is the mean author count per paper. All R^2 on $\log(1+x)$-transformed predictors.

Table 14. Sole-predictor R^2 on NameRank and marginal R^2 when added to h-index alone. Attribution-weighted measures (fractional, first/last) do not approach h-index, and add almost nothing on top of it; author count carries no signal. The count of distinct works (n_{works}) is the closest cousin of h-index.

Predictor	Sole R^2	ΔR^2 added to h-index
log(h-index)	**0.398**	—
log(n_{works})	0.337	+0.0003
log(first/last-author citations)	0.268	+0.0268
log(fractional citations)	0.154	+0.0026
log(raw citations)	0.137	+0.0001
log(mean authors/paper)	0.0004	+0.0008

In a joint OLS over all six (total $R^2 = 0.428$), the standardized coefficients are dominated by log(h-index) (0.579), with first/last-author citations (0.202) a distant second and raw citations *negative* (−0.143); fractional citations (0.109) and mean authors (0.060) are minor. The mechanism is name-recurrence across distinct works, not attribution-per-paper weighting: if the latter drove the result, fractional and first/last-author citations would match or exceed h-index, and they do not.

L Cross-Judge Robustness

A 615-record stratified sample (Section 6.7.6) was re-graded by Claude Opus 4.6 and GPT-5.5 alongside the primary Gemini 3 Flash Preview judge, holding gold answers and judge prompt fixed.

Table 15. Pairwise Pearson correlation between judges, by cohort. Agreement is high everywhere except the $n = 15$ `oss_project` cell (generic-description disagreement). The `reference_pilot` anchors—including its four Chinese-name entities—show the tightest agreement, refuting a "Chinese name → judge disagreement" hypothesis.

Subset	n	Gem–Claude	Gem–GPT-5	Claude–GPT-5
ALL	615	0.868	0.893	0.934
reference_pilot	185	0.952	0.941	0.972
(Chinese-name subset)	74	0.957	0.945	0.970
long_tail_researcher_openalex	100	0.855	0.920	0.935
cs_faculty	100	0.712	0.894	0.781
imo_gold	100	0.785	0.808	0.924
gpt5_system_card_author	100	0.781	0.917	0.830
mid_tier_writer	15	0.872	0.853	0.878
oss_project	15	0.103	0.455	0.486

Table 16. Capability-controlled in-family lift. For each response, a judge's score is compared to the mean of the *other two* judges on that same response; the lift is the own-family residual minus the other-family residual. Gemini and GPT-5 modestly favor their own family; Claude does not.

Judge	In-family residual	Other-family residual	Lift
Gemini	+0.014	−0.048	**+0.062**
GPT-5	+0.060	+0.019	**+0.041**
Claude	+0.012	+0.022	**−0.010**

Cohort-mean NameRank is stable across judges (e.g. `gpt5_system_card_author` 0.087/0.165/0.123 under Gemini/Claude/GPT-5; `long_tail_researcher_openalex` 0.669/0.630/0.631), and the silent→discriminative→universal ladder is preserved under all three. The headline findings are not a single-judge artifact.

M Synthetic-Null Floor

Thirty fictional entities (ungoogleable names verified to surface no notable real person; cohort-matched contexts and gold answers) were run through the full pipeline (Section 6.7.7). The per-archetype floor:

Table 17. Synthetic-null mean NameRank by archetype. Dense-gold archetypes (founder, CS faculty) floor near the main-run silent level; the short-boilerplate-gold IMO archetype floors much higher, because its generic year+country+medal gold is partly satisfiable from the disambiguating context without recognizing the (nonexistent) person.

Synthetic archetype	n	Mean NameRank	Refusal
synthetic_imo_gold	6	**0.199**	21%
synthetic_mid_tier_chef	1	0.123	27%
synthetic_mid_tier_journalist	1	0.092	27%
synthetic_mid_tier_podcast	1	0.075	32%
synthetic_oss_project	4	0.065	39%
synthetic_mid_tier_musician	1	0.057	41%
synthetic_cs_faculty	10	**0.027**	45%
synthetic_founder	6	**0.012**	47%
All synthetics	30	0.072	—
All except IMO archetype	24	**0.040**	—

The floor is concentrated in small open-weights fluent hallucinators: gemma-3-4b (0.293), gemma-3-12b (0.287), and llama-4-maverick (0.259) lead, while 15 of the 37 models score the synthetics at exactly 0 (including grok-4, kimi-k2, qwen3-235b, mistral-large, gemini-3.1-pro, and gpt-5.3, the last two by refusing on 100% of synthetics). Recommended use: treat ~ 0.04 as the noise floor for dense-gold cohorts and ~ 0.20 for short-boilerplate-gold cohorts; differences below the relevant floor are not evidence of recognition.

N Confound Checks: Gold Length, Context, and Wikipedia

Gold-answer length. Within-cohort regressions of NameRank on gold-answer length give $R^2 < 0.07$ for IMO (0.067), IOI (0.018), and MSRA (0.006); the high-R^2 cohorts are fame-graded mid-tier categories (actor 0.95, writer 0.88) where longer golds are an outcome of fame, not a measurement artifact. The credential-vs-OpenAlex ordering survives a pooled fixed-effects length adjustment (which deepens the gap: pooled slope $b_{\text{chars}} = -1.1\times10^{-3}$) and a within-OpenAlex-slope adjustment. A strict $|\Delta\text{chars}| \leq 50$ matched-pairs test yields $n = 0$ pairs: IMO golds span 359–392 characters, OpenAlex golds 662–789, with no overlap. The closest feasible match (IMO vs. CMO, which do overlap in length) preserves the within-credential ordering.

Disambiguating context. A 288-entity subset re-probed with the specific context replaced by a minimal generic descriptor:

Table 18. Minimal-context ablation. Stripping the specific disambiguating context to a generic descriptor drops absolute NameRank substantially but preserves the cross-entity gradient.

Comparison	Specific context	Minimal context
long_tail_researcher_openalex (mean)	0.446	0.077
imo_gold (mean)	0.345	0.039
Stanford mean	0.565	0.355
Tsinghua mean	0.258	0.062
Stanford−Tsinghua gap	**0.308**	**0.293** (−4.8%)

Absolute levels fall sharply (the model cannot identify which researcher without the context), but the Stanford–Tsinghua institutional gap shrinks by only $\sim 5\%$: context anchors disambiguation, it does not leak the gold answer.

Wikipedia presence. On the $n = 771$ long-tail cohort, a binary has-Wikipedia flag explains $R^2 = 0.022$ of NameRank (log-sitelinks and log-pageviews add +0.007 and 0.000 on top), versus $R^2 = 0.398$ for $\log(h\text{-index})$ alone. Adding log(h-index) to the full Wikipedia block lifts R^2 from 0.029 to 0.405—a marginal R^2 of 0.376. Across the full 5,719-entity cohort, Wikipedia presence explains $\sim 8\%$ of variance and cohort identity $\sim 47\%$. Wikipedia presence accounts for only $\sim 1/4$ of the Stanford–Tsinghua gap. NameRank tracks bibliometric productivity, not a Wikipedia-yes/no flag.

O Gender Gap by Cohort

First-name-inferred gender (via `gender-guesser`; ambiguous and unisex names dropped) for the person cohorts, restricted to cohorts with ≥ 5 man- and woman-coded names. The man-minus-woman NameRank gap:

Table 19. Man-minus-woman NameRank gap by cohort. The controlled researcher cohorts (top) show a small reproducible man-coded advantage that survives institution- and h-index-matching. The cross-profession cohorts (bottom) show much larger raw gaps that reflect real-world fame differences in the sampled individuals, not a NameRank artifact—hence the bias claim is restricted to the controlled researcher comparison.

Cohort	n_{man}	n_{woman}	Δ (man−woman)	p
cs_faculty	354	97	+0.038	0.014
institution-matched (60 pairs)	—	—	+0.054	0.042
long_tail_researcher_openalex	388	76	+0.031	0.182
h-index-decile-adjusted	—	—	+0.032	—
long_tail_researcher_ikp	56	22	+0.044	0.015
mid_tier_actor	38	63	+0.341	$< 10^{-3}$
mid_tier_athlete	64	32	+0.254	$< 10^{-3}$
mid_tier_writer	62	77	+0.041	0.41
mid_tier_comedian	18	17	−0.045	0.23

The within-researcher-cohort gap (+0.027 to +0.054, ∼6–8% relative attenuation) is roughly 4× smaller than the ∼2× East-Asian-surname attenuation reported by Li [2026], but reproducible across three researcher cohorts and robust to dropping medium-confidence gender labels. The large cross-profession gaps (actor, athlete) are excluded from the bias claim because the sampled men and women in those cohorts differ in actual fame.

Name-coding caveat. `gender-guesser` infers gender from first names against a predominantly Western-name dictionary; it returns "unknown" or "andy" (androgynous) for most Chinese, Korean, and other romanized East-Asian given names, which are then dropped. The gendered sub-sample is therefore enriched for Western names, and the residual gender gap is partly entangled with the East-Asian-surname attenuation it sits beside—a name that codes as gender-unknown also tends to be one of the low-recognition non-Anglophone names. The n_{woman} counts (e.g. 76–97 across the researcher cohorts) reflect this attrition. We report the gap as an inherited corpus-and-judge property restricted to the controlled within-cohort comparison, not as a clean estimate of a pure gender effect net of name origin; disentangling the two would require a gender labeling that is accurate on CJK names (e.g. manual coding or a name-origin-aware classifier).

P Limitations Summary Table

Table 20. Compact summary of methodological caveats, their direction (could overstate or understate findings), and mitigations available for future runs.

Caveat	Direction	Mitigation
Single-probe-per-entity; absolute levels are wording-conditional (Appendix I)	Shifts cardinal levels ∼0.10–0.20; rankings unaffected	*Tested:* four-template re-probe shows ordering robust ($r = 0.93$–0.98); report values as T0-conditional and hold wording fixed across studies
Judge family-bias (Appendix L)	Gemini-judging-Gemini lift +0.062 inflates per-vendor score tables; cross-entity findings unaffected	*Tested:* 3-judge re-grade ($n = 615$); report per-vendor tables under Claude or as a 3-judge mean
Cohort labels not fully orthogonal	Overcounts entities in cross-cohort comparisons	De-duplicate by name within long-tail academic categories
Chinese-prompt context fields remained English	Understates cross-language effect	Pure-Chinese prompts for the full 240-entity sub-run
Single judge on the full run (Gemini 3 Flash Preview)	Family-specific scoring style on per-vendor tables	*Tested:* non-Gemini replication (Claude, GPT-5) on $n = 615$; a full multi-judge consensus run would remove the offset
Cross-section, not longitudinal	Cannot infer name-propagation rates	Re-run at each major model-release cycle; release as NameRank v2, v3, ...
Scores not comparable across model vintages (Appendix J)	∼0.13/yr capability drift inflates any cross-fleet level comparison	*Tested:* silent zone is intrinsic, not timing (matched DiD ≈0); make longitudinal claims on within-run relative gaps only
Researcher mobility confound in country gradient	Inflates US averages, deflates China averages	Re-classify by country-of-PhD or country-of-doctorate-training as alternative slicing
Tsinghua/Peking small samples ($n = 4$)	Institution-level reads noisy at the bottom	Expand sample by drawing from CCF rank-A faculty rosters
Tianshou case study sample of 1 for per-response mediation analysis	Cannot generalize the +0.36 score lift cleanly	Extend per-response mediation analysis to all 11 verified (creator, artifact) pairs
Inherited gender bias (Appendix O)	Man-coded researcher names score +0.03–0.05 over woman-coded	*Measured, not corrected:* report alongside the East-Asian-surname effect; restrict to controlled within-cohort comparisons
Short-boilerplate-gold cohorts have an elevated noise floor (Appendix M)	Olympiad-credential scores up to ∼ 0.20 are partly context-prior leakage	Read scores against the synthetic floor (∼ 0.04 dense-gold, ∼ 0.20 short-gold); the floor widens, not narrows, the credential gap
Institution variance share inflated by category count (Section 6.4.2)	Naive $R^2 = 0.54$ counts 319 dummies on 698 points; true share ∼16%	*Corrected:* report adjusted R^2/ω^2/ICC (∼0.16) and the large-cell Stanford–Tsinghua contrast; country-level aggregation is not inflated
Context-injection lift includes a coverage-echo component (Section 6.3.4)	Overstates the named-artifact-amplification effect; lift is an upper bound	Re-judge arm B against an artifact-redacted gold crediting only non-injected facts; report the residual lift
Gold/training-source overlap for Wikipedia-derived golds (Section 4)	For ∼60% of entities, coverage partly measures Wikipedia memorization	*Tested:* a binary Wikipedia flag explains only ∼2% of long-tail variance vs. 40% for h-index (Appendix N)